THE MARKETPLACE

Enterprise Data Products & AI Agents on One Governed Shelf

Product Blueprint & Build Specification

Version 2.0 · September 2026 · Vendor-Neutral · Governed by Construction · Demo-Ready

What this document is

A complete, build-ready specification for a multi-industry marketplace in which data products and the AI agents that run on them are catalogued, governed, requested, observed, and demonstrated as a single supply chain.

It is written to be handed directly to a coding agent (Claude Code / Cortex Code) or an engineering team. Every rubric, score, contract, and mesh edge in this specification is stored as data, never hardcoded.

Seed content covers 15 domain data products and 14 AI agents across Telecom, Technology, Banking, Insurance, Healthcare, Retail, Transportation, Utilities & Energy, and Manufacturing — sufficient for a live client demonstration on day one.

Attribute	Value
Product name	Deliberately unbranded. The display name, wordmark and palette are theme configuration (PRODUCT_NAME token); no code path contains a brand string.
Document type	Product blueprint + implementation specification
Companion asset	EAIRN — Enterprise AI Readiness Navigator (assessment upstream of marketplace onboarding)
Primary platform target	Snowflake-native (Cortex Analyst, Cortex Search, Cortex Agents, Snowflake Intelligence); portable to Databricks, Fabric, BigQuery
Consumption surfaces	Web portal, REST API, MCP server endpoint, SQL / semantic view, notebook SDK
Non-negotiables	Rubrics as data · metadata-only by default · every inference carries confidence + rationale · human approval before promotion · immutable audit snapshots

Contents

1. Executive Summary

Enterprises have spent a decade building data platforms and are now spending a year building AI agents on top of them. The two efforts are almost never connected. Data products are published into catalogs that business users do not visit; agents are built in isolation by teams who cannot see which certified data product already answers the question they are hand-rolling a pipeline for. The result is duplicated supply, ungoverned agent access, and a consumer population that still files a ticket and waits three weeks.

The Marketplace is an enterprise-grade platform that treats data products and AI agents as one governed supply chain. A business user browses domain data products, sees the agents that already run on them, watches those agents answer real questions in a live demo console, requests access in a workflow that provisions entitlements mechanically, and — when nothing in the catalog fits — files a demand request that lands in the same backlog that produced everything else on the shelf.

1.1 Seven Capabilities That Define the Product

Capability	What it delivers
Dual catalog	Data products and AI agents are first-class, cross-linked citizens of one catalog. Every agent declares the data products it consumes; every data product exposes the agents that consume it.
Evidence-backed listings	Every listing carries a live data quality score, freshness posture, owner, data contract, consumption endpoints, adoption metrics, and a quantified value case — sourced from telemetry, not from a wiki page.
Live demo console	Each agent ships with a minimum of five curated question–answer exchanges that execute against real (or demo-tier) data, so a prospective consumer sees the answer before requesting access.
Governed request rails	Three request types — access, enhancement, and new-supply demand — run through one workflow engine with SLA clocks, approval chains, and mechanical provisioning on approval.
Two meshes	A data product mesh linking products that share upstream sources, and an agent mesh linking agents by KPI, domain and data-product overlap. Both are computed from lineage and metadata, not drawn by hand.
Observability plane	Freshness, quality, contract conformance, cost, latency, groundedness, deflection and drift — for products and agents alike — with regression alerting to owners.
Academy	Role-based learning paths, certification, and an in-context "how do I consume this" runbook attached to every listing.

1.2 Why This Is Different

A catalog tells you what exists. A marketplace makes you want it, lets you try it, and gets you access. Three design decisions carry that difference.

- **Supply and demand are modelled symmetrically.** Most catalogs model only supply. The Marketplace models unmet demand as a first-class object with the same lifecycle, scoring and visibility as a published product — so the roadmap is driven by observed consumer need rather than platform-team intuition.
- **Agents are catalog entries, not features.** An agent has an owner, a contract, a cost per answer, an accuracy posture, an entitlement scope, and a value case — exactly like a data product. Treating agents as inventory is what makes them governable at scale.
- **Trust is rendered, not asserted.** Quality scores, contract SLAs, and groundedness rates are drawn from the observability plane and displayed on the listing at browse time. A consumer never has to leave the marketplace to decide whether an asset is safe to depend on.

1.3 Scope of This Specification

Sections 1–6 define the product and its front-of-house experience. Sections 7–12 specify the catalog and detail surfaces in wireframe-level detail. Sections 13–15 define the three request workflows. Sections 16–20 cover the mesh, value analytics, observability and academy modules. Sections 21–24 give the engineering team a canonical data model, physical schema, reference architecture, API surface, and non-functional requirements. Sections 25–27 supply the complete seed catalog and 60 scripted demo exchanges. Sections 28–36 cover the disciplines a production platform organisation requires: the KPI registry, AgentOps, FinOps, standards conformance, the design system, test and release engineering, estate bootstrap, product analytics and the commercial model. Sections 37–40 give the build plan, roadmap, demonstration script and risk register.

2. Business Problem

Five failures recur across every enterprise data estate large enough to need a marketplace. Each maps to a specific capability in this specification.

2.1 The Five Failures

Failure	How it shows up	Platform response
Discovery collapse	Consumers cannot find certified data, so they rebuild it. Enterprises routinely run 4–8 competing definitions of the same KPI.	Domain-organized catalog with certified-KPI search, semantic matching, and duplicate-supply detection at intake (§7, §15)
Trust vacuum	Even when an asset is found, the consumer has no way to judge whether it is fresh, accurate or supported — so they escalate to a human for reassurance.	Quality score, freshness clock, contract SLA and incident history rendered on every card and detail page (§8, §19)
Access friction	Access requests are email threads and ticket queues. Median time-to-first-query is measured in weeks, and entitlements are granted broadly because narrow scoping is manual work.	Structured access request with purpose capture, automated policy evaluation, and mechanical least-privilege provisioning on approval (§13)
Agent sprawl	Agents proliferate outside governance. Nobody can answer which agents touch regulated data, what they can be asked, what they cost, or whether their answers are grounded.	Agents catalogued with entitlement scope, KPI coverage map, cost and groundedness telemetry, and an agent mesh that exposes duplication (§10, §17, §19)
Invisible demand	When the catalog does not serve a need, the need disappears into a spreadsheet. The platform roadmap is therefore built on anecdote.	Demand intake with vote aggregation, business-value scoring, duplicate clustering, and a public demand board (§15)

2.2 The Cost of the Status Quo

- **Duplicated supply.** Where KPI definitions are unmanaged, 30–60% of pipeline spend reproduces data that already exists in a certified form elsewhere in the estate.
- **Stalled AI programs.** Copilot and agent pilots stall at security review because the entitlement scope of the agent cannot be described, let alone constrained, and because no artifact demonstrates what the agent will and will not answer.
- **Unrealized platform investment.** A data product nobody adopts has negative return: it carries storage, compute, stewardship and on-call cost with no offsetting value. Without adoption telemetry, low-value products are never retired.
- **Consumer attrition.** Business users who wait weeks for access stop asking. Shadow extracts, unmanaged spreadsheets and personal copies of regulated data are the direct consequence of request friction.

2.3 What Good Looks Like

Target-state outcomes the product is designed to produce

Time from discovery to first successful query: under one business day for pre-approved tiers, under three days for regulated tiers.

Share of analytical questions answered by a catalogued agent rather than a human analyst: above 40% within four quarters.

Share of new pipeline requests that intake identifies as duplicate supply and redirects to an existing product: above 25%.

Every regulated data element consumed by an agent traceable to an approved entitlement grant and a logged purpose.

Every published asset carrying a current quality score, a named owner, and a value case reviewed within the last two quarters.

3. Product Vision & Design Principles

3.1 Vision Statement

> *Any employee should be able to find a governed data product or AI agent for their question in under two minutes, see it answer that question live before committing to it, and be querying it themselves within a day — with every access, answer and dollar of value fully attributable.*

3.2 Design Principles

These principles are enforced in code review and in the CI test suite. A pull request that violates one is rejected regardless of feature value.

#	Principle	Enforcement
1	Everything scored is stored as data	Quality rubrics, mesh similarity weights, value-scoring models, grade bands and SLA thresholds live in versioned configuration tables. CI test asserts no numeric threshold appears in application source.
2	Metadata-only by default	Marketplace surfaces operate on metadata, contracts and aggregate telemetry. Row-level preview is a separate permission-gated module with per-dataset authorization and access logging.
3	Tool-neutral source of truth	Every listing derives from a YAML product/agent manifest. Platform artifacts (Snowflake semantic views, dbt models, MCP tool definitions, LookML) are generated renderings, header-stamped as auto-generated.
4	Every inference carries confidence and rationale	Mesh edges, duplicate-supply detection, value estimates and auto-generated descriptions store a confidence score and a rationale. Below 0.80 they are routed for human review before display.
5	Human approval before promotion	No agent, data product, contract version or entitlement grant reaches production without a recorded human approval, actor identity and timestamp.
6	Classification propagates mechanically	PII and sensitivity tags propagate from source through the product contract into agent tool scopes and retrieval filters. An agent cannot be published over a product carrying tags its scope does not cover.
7	Immutable audit snapshots	Every published version of a product, agent, contract or score is an immutable, keyed snapshot. Historical listings are replayable exactly as a consumer saw them.
8	Demo parity with production	The demo tier runs the same code path as production against a synthetic dataset. There are no mocked answers, and demo mode is visibly labelled.

3.3 What This Product Is Not

- Not a data catalog. It integrates with Collibra, Alation, Purview, Atlan, DataHub and Unity Catalog as evidence sources and pushes certified listings back to them.
- Not an ETL or transformation tool. It publishes and governs the products that pipelines produce; it does not build the pipelines.
- Not an agent development framework. Agents are authored in Cortex Agents, Bedrock, LangGraph or equivalent; the Marketplace catalogues, scopes, meters, observes and demonstrates them.
- Not an identity provider. It orchestrates entitlement requests and provisions through the platform’s native RBAC and the enterprise IdP; it never becomes a shadow authorization store.

3.4 Federated Operating Model

A marketplace is an operating model before it is software. The platform assumes federated domain ownership: domains own supply, a central platform team owns the rails, and a governance forum owns the rules. Where an organisation does not yet operate this way, the platform makes the gap visible rather than hiding it — an asset with no accountable domain owner cannot be certified.

Body	Owns	Decision rights	Held accountable for
Domain team	Data products and agents in its domain	Contract content, schema evolution, enhancement backlog, access approval for its assets	Quality composite, contract conformance, adoption and value case of its assets
Platform team	The marketplace, connectors, generators, workflow and observability rails	Canonical model, generator conventions, non-functional budgets, connector permission manifests	Availability, request latency, provisioning correctness, platform cost per asset
Governance forum	Rubrics, taxonomies, certification standard, KPI register	Rubric versions and weights, certification criteria, purpose categories, sensitivity policy	Consistency of scores across domains and the integrity of the KPI register
Security & privacy	Entitlement policy and agent scope standards	Approval thresholds by tier, residency rules, agent autonomy ceilings	Reconstructability of any access; zero unscoped agent identities
Executive sponsor	Portfolio investment	Funding, retirement decisions, demand prioritisation at portfolio level	Portfolio net value and unmet-demand throughput

- **Rubric changes are a governance act, not a deploy.** Weights live in versioned tables, but a new rubric version requires forum approval and re-scores the estate under the new version while preserving prior snapshots. Scores are never silently restated.
- **Certification is federated, standardised centrally.** Domain stewards certify; the criteria they certify against are a single central rubric. A domain cannot invent a softer bar for itself.
- **The platform team does not own content.** It owns the fact that content cannot be published without an owner, a contract, a limitation statement and a value case. Enforcement lives in the publish gate, not in a review meeting.

4. Personas, Journeys & Access Model

4.1 Personas

Persona	Primary goal	Key surfaces	Success signal
Business Consumer	Answer a business question without writing SQL or waiting on an analyst	Landing page, catalog, agent demo console, access request	Question answered in-session; access granted same day
Data Analyst / Scientist	Find trustworthy, contract-backed inputs for a model or analysis	Product detail, schema explorer, contract tab, endpoints, mesh	Reuses an existing product instead of building a pipeline
Data Product Owner	Grow adoption, defend investment, manage the roadmap of their product	Owner console, adoption analytics, enhancement backlog, observability	Adoption and value trends up; contract breaches at zero
Agent Owner	Ship a safe, accurate, in-demand agent and prove its deflection value	Agent studio, KPI coverage editor, demo curator, groundedness dashboard	Answer acceptance rate above target; deflection hours rising
Data Steward	Ensure classification, ownership and contract conformance across the domain	Certification queue, low-confidence review queue, policy conformance view	Certification backlog cleared within SLA
Platform / Domain Architect	Reduce duplication, plan capacity, govern the mesh	Mesh explorer, duplicate-supply report, cost analytics	Duplicate supply detected pre-build; mesh density improving
Security & Privacy Officer	Prove that every agent and consumer access is scoped, purposed and auditable	Entitlement register, purpose log, agent scope report, audit export	Complete access reconstruction for any answer on demand
Executive Sponsor (CDO/CDAO)	See portfolio value, adoption and unmet demand in one place	Executive dashboard, value realization board, demand board	Portfolio value case defensible at budget review

4.2 The Signature Journey — Question to Query in One Day

1. **Arrive.** A regional sales director lands on the front page and types "why is churn up in the Midwest" into the universal search bar. Search resolves across product names, KPI definitions, agent capability statements and glossary terms.
2. **Recognize.** Results return the Subscriber Churn & Retention 360 data product (quality 94, certified, 340 monthly consumers) and the Churn Sentinel Agent, ranked by semantic match against the KPI "churn rate" rather than by keyword.
3. **Try before requesting.** The director opens the agent, plays scripted question 3 ("Which segments drove the churn increase last quarter?"), and sees a live answer with a chart, the KPI definition used, and citations to the underlying data product columns.
4. **Ask their own question.** In demo tier, the agent answers against synthetic data and clearly labels it as such — enough to confirm fit without granting production access.
5. **Request access.** One click opens a pre-filled request: assets, purpose, duration, sensitivity acknowledgement. Policy evaluation runs immediately and shows which approvals are required and the SLA clock.
6. **Approve and provision.** The product owner approves in the console or by email action. Entitlements are provisioned mechanically to the platform role, the agent scope is extended, and the consumer is notified with a runbook.

7. **Consume and be counted.** First query is logged against the product, attributed to the consumer and purpose, and rolled into adoption and value telemetry visible to the owner the next morning.

4.3 Access Model

Six roles, evaluated against asset sensitivity tier. Marketplace roles govern the marketplace; data access itself is always provisioned into the platform's native RBAC.

Marketplace role	Catalog browse	Demo console	Request	Approve	Publish	Admin
Viewer	Public + internal tiers	Demo tier only	Yes	No	No	No
Consumer	All tiers (metadata)	Demo + granted assets	Yes	No	No	No
Owner	All tiers	Own assets, full	Yes	Own assets	Own assets	No
Steward	All tiers	Domain assets	Yes	Domain certification	Certify only	No
Architect	All tiers	All (read)	Yes	Mesh & intake triage	No	Mesh config
Administrator	All tiers	All	Yes	Break-glass	All	Full

Sensitivity tiers are Public, Internal, Confidential and Restricted. Restricted assets never render sample values, never appear in demo tier with real data, and always require both owner and privacy approval. The tier is inherited from the highest classification present in the product contract and cannot be lowered in the marketplace.

5. Information Architecture & Navigation

5.1 Top-Level Structure

```
/                          Landing experience (unauthenticated + personalized)
/discover                  Universal search across products, agents, KPIs, glossary
/data-products             Catalog: filter, facet, compare
/data-products/:id         Detail — 8 tabs (Overview, Schema, Quality, Contract,
                           Endpoints, Lineage & Mesh, Consumption, Value)
/agents                    Agent catalog: filter by domain, KPI, data product
/agents/:id                Detail — 7 tabs (Overview, Capabilities, Demo, Data
                           Products, Safety & Scope, Observability, Value)
/agents/:id/demo           Full-screen live demo console
/mesh/data                 Data product mesh explorer
/mesh/agents               Agent mesh explorer
/requests                  My requests · Approvals queue · SLA board
/requests/new/access       Access request
/requests/new/enhancement  Enhancement request
/requests/new/supply       New data product / agent demand intake
/demand                    Public demand board with voting and clustering
/observability             Health plane for products and agents
/academy                   Learning paths, certifications, runbooks
/console/owner             Owner workspace: adoption, backlog, contract, incidents
/console/steward           Certification and low-confidence review queues
/admin                     Rubrics, weights, taxonomies, connectors, tenancy
```

5.2 Global Navigation

- **Persistent header** — logo, universal search (Cmd/Ctrl-K), primary nav (Discover · Data Products · Agents · Mesh · Observability · Academy), request counter, notifications, avatar menu.
- **Universal search** is the primary navigation instrument. It searches four indexes simultaneously — asset names and descriptions, certified KPI definitions, agent capability statements, and business glossary terms — and returns typed, grouped results with inline quality and adoption badges.
- **Breadcrumbs** on every detail page reflect the domain taxonomy: Industry › Domain › Sub-domain › Asset.
- **Contextual right rail** on detail pages carries the persistent action stack: Request Access, Try in Demo, Request Enhancement, Add to Comparison, Subscribe to Changes, Copy Endpoint.

5.3 Taxonomy

Three orthogonal taxonomies classify every asset. All three are stored as versioned reference data and are editable in /admin without a release.

Taxonomy	Values	Purpose
Industry	Telecommunications · Technology & Software · Banking & Capital Markets · Insurance · Healthcare & Life Sciences · Retail & Consumer · Transportation & Logistics · Utilities & Energy · Manufacturing · Public Sector · Media & Entertainment · Hospitality & Travel	Vertical filter and demo-persona switching
Business domain	Customer · Product · Finance · Risk & Compliance · Operations · Supply Chain · Workforce · Marketing · Network & Asset · Clinical	Mesh clustering and ownership alignment
Data product	Source-aligned · Aggregate · Consumer-aligned · Feature · Document corpus ·	Determines applicable quality

Taxonomy	Values	Purpose
archetype	Reference / master data	dimensions and contract template

6. Landing Experience — Front Page Design Specification

The front page has one job: convince a first-time visitor within eight seconds that this marketplace contains something they need and that it is trustworthy. It is the primary sales surface in a client demonstration, so it is specified here at pixel-intent level.

6.1 Visual System

Token	Value	Applied to
Deep Navy	#0B2C4D	Hero background base, headings, primary text
Teal	#117A8B	Accent gradient stop, section headers, active states, links
Signal Gold	#B98A1B	Primary CTA, certification marks, emphasis rules
Surface	#FFFFFF / #F5F9FA	Cards, page background, alternating table rows
Ink / Muted	#1F2A33 / #4A5C68	Body copy, secondary metadata
Status	Green #1E7F4F · Amber #B57614 · Red #A62828	Quality bands, SLA state, incident severity
Ocean gradient	linear-gradient(135deg, #0B2C4D 0%, #117A8B 62%, #1AA3A3 100%)	Hero canvas, mesh node glow, empty-state illustrations
Type	Display: Inter Tight 600/700 · Body: Inter 400/500 · Mono: JetBrains Mono	Hero 56/64px, H1 40px, H2 28px, body 16px, caption 13px
Geometry	Radius 12px cards / 8px controls · Shadow 0 1px 2px + 0 8px 24px rgba(11,44,77,.08) · 8px spacing grid	All surfaces
Motion	160ms ease-out hover, 320ms card entrance, 600ms mesh settle; all suppressed under prefers-reduced-motion	Interactive elements

6.2 Page Composition — Ten Bands

#	Band	Content specification
1	Hero	Ocean-gradient canvas with a slowly animating mesh constellation (nodes = data products, edges = shared sources) rendered at 12% opacity behind the copy. Headline: "Every governed data product and AI agent in the enterprise. One shelf." Sub-headline names the value: find it, watch it answer, request it, use it today. Centre-stage universal search bar with rotating placeholder examples drawn from the live catalog. Two CTAs: Browse the catalog (gold, filled) and Watch an agent answer (ghost).
2	Trust strip	Four live counters read from the platform, not hardcoded: data products published, AI agents live, certified KPIs, questions answered this month. Each animates on scroll and links to the corresponding filtered view.
3	Industry selector	Horizontally scrollable row of 12 industry tiles with icon, product count and agent count. Selecting a tile re-renders bands 4–6 for that vertical — this is the demo-persona switch used to tailor a client walkthrough in one click.
4	Featured data products	Three-across card grid, six cards. Each card: name, domain chip, one-line purpose, quality score ring, freshness pill, consumer count, certification mark, three most-used KPIs. Cards are selected by a ranking function (adoption velocity × quality × recency) stored as rubric data.

#	Band	Content specification
5	Meet the agents	Carousel of agent cards, each with avatar glyph, capability statement, the data products it runs on, and a "Try a question" button that opens the demo console inline without leaving the page. This band is the conversion engine of the front page.
6	Live answer theatre	A single embedded demo panel that auto-plays one scripted exchange (typed question → streaming answer → chart → citations) on a 20-second loop, with controls to pause and to choose a different question. Demonstrates capability without requiring a click.
7	How it works	Four-step horizontal timeline — Discover · Try · Request · Consume — each step with a 40-word explanation and a screenshot thumbnail. Anchors the mental model for first-time visitors.
8	The mesh	Interactive miniature of the data product mesh; hovering a node reveals its neighbours and shared sources. Click-through to /mesh/data. This band is the visual differentiator against ordinary catalogs.
9	Value proof	Three quantified outcome tiles fed from the value module (analyst hours deflected, duplicate builds avoided, median time-to-access) plus two customer-quote slots that render placeholder copy until populated.
10	Footer	Domain sitemap, governance and privacy statements, connector status, API and MCP documentation links, academy entry point, support and SLA contacts.

6.3 Personalization Rules

- Unauthenticated visitors see the full marketing composition with demo-tier data only.
- Authenticated users see bands 4 and 5 re-ranked by their department’s consumption history and by assets their peers in the same organizational unit adopted in the last 90 days, with a "Because your team uses..." caption stating the reason. The caption is mandatory — no unexplained recommendations.
- Users with pending requests see a status strip above band 2 with the SLA clock.
- Owners see an inline strip with their assets’ health and any breach requiring attention.
- A visitor’s industry is inferred from tenant configuration; the inference is overridable in one click and the override persists.

6.4 Performance & Accessibility Budget

Metric	Budget
Largest Contentful Paint	≤ 1.8 s on a 4G connection; hero renders from server components with no client-side data dependency
Total blocking time	≤ 150 ms; mesh canvas and answer theatre lazy-load below the fold
Interaction to next paint	≤ 200 ms for search-as-you-type
Accessibility	WCAG 2.2 AA. Contrast ≥ 4.5:1 for all text including gold-on-navy CTAs; full keyboard traversal; mesh has an equivalent accessible table view; all animation respects prefers-reduced-motion
Internationalization	All copy externalized; layout tested for 35% string expansion; date, number and currency formatting locale-aware

6.5 Motion System — Principles and Tokens

The front page moves because the estate is alive, not because motion is fashionable. Every animation on this page answers one of three questions: what is on the shelf right now, what can it do, and is it working. Motion that answers none of them is removed.

Motion class	Definition	Where used	Rules
Ambient	Continuous, low-salience, never demands attention or blocks reading	Hero mesh constellation, data product ribbon, agent orbits	Amplitude under 8px, opacity under 0.3 for background layers, pausable, always secondary to text contrast
Reactive	Triggered by pointer, focus, scroll or keyboard; ends when input ends	Card hover lift, node neighbourhood highlight, counter count-up on scroll, ribbon pause on hover	Under 200ms to first visible response; must be reachable by keyboard, not pointer only
Narrative	Plays once to tell a specific story, with explicit user controls	Live answer theatre, how-it-works timeline, mesh settle on first paint	Pause, restart and skip controls; never auto-restarts more than once without interaction

Token	Value	Applied to
duration.micro / base / slow / ambient	120ms / 240ms / 480ms / 20–90s loop	Hover feedback · card entrance · mesh settle · ribbon and orbit cycles
easing.standard / entrance / exit	cubic-bezier(.2,0,0,1) · cubic-bezier(.05,.7,.1,1) · cubic-bezier(.3,0,.8,.15)	All transitions; no linear easing except continuous marquee translation
stagger.grid / list	40ms per card, capped at 240ms total · 24ms per row	Card grids and result lists on first paint only, never on filter re-render
animatable properties	transform and opacity only	Every animation. Animating width, height, top, left, box-shadow or filter is a review rejection
motion.pauseTriggers	prefers-reduced-motion · document.hidden · Save-Data header · battery below 20% · viewport under 640px · user pause control	Global motion controller; any trigger halts ambient motion and freezes to a composed static frame

6.6 Living Data Product Ribbon

Band 4 renders as a continuously moving supply ribbon rather than a static grid, so a first-time visitor sees breadth in the first three seconds without scrolling or clicking. It is the single most persuasive element on the page and the one most likely to be built badly, so it is specified precisely.

Aspect	Specification
Composition	Two rows of data product cards travelling in opposite directions. Upper row right-to-left at 32px/s, lower row left-to-right at 24px/s. Different speeds prevent the two rows from reading as one moving block.
Card content	The standard catalog card (§7.3) at 88% scale: name, domain chip, one-line purpose, animated quality ring, freshness pill, consumer count, attached-agent avatars.
Selection	Live from GET /products?featured=true, ordered by the featured ranking function (adoption velocity × quality × recency) stored as rubric data. Minimum 12 and maximum 24 cards per row; the set is re-fetched every 5 minutes and cross-faded, never hard-swapped.

Aspect	Specification
Loop mechanics	Track duplicated once and translated with transform: translate3d(); the transform resets at exactly one track width so the seam is never visible. Implemented with the Web Animations API on a single composited layer per row — not one animation per card, and not a scroll-position hack.
Interaction	Pointer enter or keyboard focus anywhere in a row pauses that row within 120ms and lifts the hovered card 4px with a shadow step. Leaving resumes from the paused position, never from the start. Click or Enter opens the product detail page.
Keyboard and assistive tech	The ribbon is a semantic list. Tab moves card to card in DOM order and pauses motion for as long as focus is inside. A visible Pause / Play control sits at the band's top-right and satisfies WCAG 2.2.2 for motion lasting longer than five seconds.
Quality ring animation	On card entry into the viewport the ring sweeps from 0 to its score over 700ms with the numeric value counting in parallel, then holds. It does not re-animate on each loop pass — repeated animation of the same number reads as instability.
Reduced motion	Ribbon renders as a static three-across grid of the top six cards with a "Browse all 15 products" link. No horizontal auto-scroll, no count-up, no cross-fade.
Empty and error state	If the catalog API fails the band renders the last successful payload from cache with an as-of stamp; if there is no cache it collapses entirely rather than showing skeletons on a marketing page.

6.7 Agent Constellation and Hero Mesh

The hero background renders the estate itself: data products as nodes, shared upstream sources as edges, and agents as satellites in slow orbit around the products they consume. It is the same data as the mesh explorer in Section 16, rendered at low salience, and it is what makes the page unmistakably a data and AI product rather than a generic SaaS landing page.

Aspect	Specification
Data source	GET /mesh/data?scope=featured returns up to 60 product nodes and their edges above the render threshold, plus GET /agents?live=true for satellites. Real identifiers, real edges — the hero is never a decorative fake graph.
Layout	d3-force simulation pre-warmed for 300 ticks server-side and shipped as fixed coordinates, so the client paints the settled layout immediately. The client then runs a damped simulation at alpha 0.02 for gentle drift only.
Agent orbits	Each agent renders as a small glyph orbiting the centroid of the products it consumes, at 0.6–1.1 degrees per second with per-agent phase offset. An agent consuming two products traces an ellipse between them, which makes multi-product agents visually legible without a legend.
Answer pulses	When an agent answers a question, a pulse travels along the agent-to-product edge in 900ms. Fed by a Server-Sent Events channel of anonymised, aggregated activity; in demo tier it replays a recorded event stream. Rate-limited to six pulses per second so the hero never strobes.
Encoding	Node radius = active consumers (log scale, clamped 3–14px); node colour = quality band; ring = certification; slow amber pulse = open incident. The same encoding as the full mesh explorer, so the visual vocabulary is learned once.
Interaction	Hover or focus freezes drift, dims non-neighbours to 20% opacity, and shows a compact peek card with name, quality, consumers and attached agents. Click navigates to the asset. The hero is interactive but never captures scroll.
Rendering technology	SVG below 150 nodes; a single canvas layer with quadtree hit-testing above that. WebGL only if a tenant estate exceeds 1,500 rendered nodes, which the featured scope should prevent. Rendering runs off the main thread where

Aspect	Specification
	OffscreenCanvas is available.
Level of detail	Below 900px viewport width the constellation drops edges and renders nodes only. Below 640px it is replaced by a static gradient with a single composed graph image.
Text protection	The constellation is capped at 12% opacity behind copy and a navy scrim guarantees 4.5:1 contrast for hero text regardless of node density. Contrast is asserted by an automated check in CI against the rendered hero.
Reduced motion	A single settled frame renders with no drift, no orbit and no pulses. Hover highlighting remains, because it is input-driven rather than ambient.

6.8 Live Answer Theatre

Band 6 shows an agent answering a real question, unattended, on a loop. It converts better than any static screenshot because it demonstrates the product doing the thing the product is for.

- **Choreography.** Question types character by character at 45 characters per second with a blinking caret (900ms) → 400ms thinking indicator showing the tool being called → answer headline streams token by token → chart draws in over 600ms with bars growing from the axis → data table rows fade in with 24ms stagger → citation chips and the as-of stamp slide up last. Total 14–18 seconds, then a 4-second hold before the next exchange.
- **Provenance is honest.** For unauthenticated visitors the theatre replays a recorded execution trace of a real agent run — the tokens, tool calls, latency and cost are exactly what that run produced — and carries a "recorded" stamp with its date. Traces are re-recorded by the nightly validation job, so a stale trace cannot survive a regression. Authenticated users see live execution against the demo tier instead. Fabricated answers are prohibited in both modes.
- **Controls.** Pause, restart, previous and next exchange, plus a question picker listing the curated set. Keyboard operable. The loop stops permanently after three cycles without interaction, because indefinite looping is an accessibility and attention cost with no additional persuasive value.
- **The right rail is visible by default.** Tool calls, rows scanned, latency and cost render alongside the answer. This transparency panel is the strongest differentiator against a generic chat demo and it is not hidden behind a toggle.
- **Reduced motion.** The completed answer renders instantly and in full, with the chart drawn and no typing effect. Every control still works.

6.9 Activity Ticker and Live Counters

- **Counters (band 2)** count up from zero to the live value over 1,200ms with entrance easing, triggered once when the strip enters the viewport. Values come from a cached aggregate refreshed every 30 seconds; a changed value tumbles digit by digit rather than jumping.
- **Activity ticker** runs beneath the counters: a slow vertical crawl of anonymised platform events — "a retail data product was certified", "an access request was provisioned in 3 hours", "an agent answered 240 questions today". Never a person’s name, never an asset above Internal sensitivity, never a customer identifier.
- **Privacy rule.** Ticker events pass through the same aggregation and suppression rules as the benchmark service: no event referencing fewer than five underlying occurrences is emitted, which prevents the ticker from becoming a side channel onto a single team’s activity.
- **Failure behaviour.** If the event channel is unavailable the ticker hides itself. A frozen ticker showing a stale event is worse than no ticker.

6.10 Motion Performance and Accessibility Budget

Metric	Budget	Enforcement
Sustained frame rate	58 fps or better on a 4-year-old mid-range laptop and on an iPhone SE class device with all ambient motion running	Playwright trace with CPU throttling in CI; regression fails the build
Main-thread work per animation frame	≤ 4ms; layout and paint excluded from animation frames entirely	Only transform and opacity animate; CI lints for animation of layout-triggering properties
Cumulative Layout Shift	0.00 on the landing page. Ribbon, constellation and theatre reserve their exact box before paint	Lighthouse assertion in CI
Interaction to next paint	≤ 200 ms with all ambient motion running, including ribbon pause on hover	Field and lab measurement; ambient motion yields to input
Off-screen and hidden cost	Zero animation frames scheduled when the element is off-screen or document.hidden	IntersectionObserver and visibilitychange wired into a single motion controller, not per-component
Reduced motion	Every ambient animation has a specified static equivalent that preserves the same information	Visual regression suite runs the full landing page in both motion modes
Motion pause control	A visible, keyboard-reachable global "Reduce motion on this page" control in the header, persisted per user	WCAG 2.2.2 conformance test
Bundle cost of motion	≤ 45 KB gzipped for the entire motion layer including the graph renderer	Bundle budget check in CI; a heavyweight animation library that breaches it is rejected

7. Data Product Catalog — Browse & Discovery

The catalog answers "what exists, is it any good, and who else uses it" before the consumer opens anything. Every judgment a consumer needs to shortlist an asset is rendered on the card.

7.1 Layout

- **Left rail (280px, collapsible):** faceted filters with live result counts.
- **Main region:** result count, sort control, view toggle (grid / table / compare), then results. Grid is three-across at ≥1440px, two at ≥1024px, one on mobile.
- **Compare mode:** select up to four products to render a side-by-side attribute matrix (quality, freshness, contract SLA, consumers, cost, agents attached, overlap with each other).
- **Empty state:** never a dead end. Zero results renders the nearest semantic matches, the top three related demand-board entries, and a primary CTA to file a new-supply request pre-filled with the query.

7.2 Facets

Facet	Values / behaviour
Industry	12 industry values from the taxonomy; multi-select
Business domain	10 domain values; multi-select
Archetype	Source-aligned, Aggregate, Consumer-aligned, Feature, Document corpus, Reference
Certification	Certified · Published · Beta · Deprecated. Deprecated hidden by default with a visible toggle
Quality score	Range slider over the 0–100 composite, with band shortcuts (≥90 Exemplary, 75–89 Healthy, 60–74 Watch, <60 At Risk)
Freshness	Real-time · Hourly · Daily · Weekly · Monthly, evaluated against the contract, not the schedule
Sensitivity	Public · Internal · Confidential · Restricted
Access status	I have access · Request required · Pre-approved for my role
Consumption surface	SQL · REST · MCP · Streaming · File · Semantic view
Agent-enabled	Boolean — has at least one live agent attached
Owner / domain team	Type-ahead over the owner directory
Adoption	Consumer-count bands and 90-day adoption trend direction

7.3 Card Anatomy

Twelve elements, fixed positions, no exceptions — consistency across cards is what makes the grid scannable.

Zone	Elements
Header	Product name (18px semibold) · certification mark · sensitivity chip
Sub-header	Industry chip · domain chip · archetype chip
Body	One-line purpose statement, hard-truncated at 120 characters

Zone	Elements
Metric row	Quality score ring (colour-banded) · freshness pill showing time since last successful load against SLA · consumer count (90-day active) · monthly query count
KPI strip	Up to three certified KPI names this product serves, as chips
Agent strip	Avatars of up to three attached agents with an overflow counter
Footer	Owner avatar and team name · last updated · primary action (Request access / Open) · secondary action (Try an agent)

7.4 Ranking & Search

Default ordering is a relevance function whose coefficients are stored in the ranking rubric table and tunable per tenant without a release:

```
score = w1 · semantic_match(query, name+purpose+kpi_definitions)
      + w2 · normalized(quality_composite)
      + w3 · normalized(active_consumers_90d)
      + w4 · certification_multiplier          -- certified 1.0, published 0.8, beta 0.5
      + w5 · peer_affinity(user.org_unit)      -- adoption by the user’s organizational unit
      - w6 · staleness_penalty(days_since_contract_review)

Defaults (rubric v1.0): w1 0.35, w2 0.20, w3 0.15, w4 0.15, w5 0.10, w6 0.05
```

Semantic matching runs over an embedding index covering product names, purpose statements, certified KPI definitions, column business names and glossary synonyms. Lexical matching runs in parallel; results are fused with reciprocal rank fusion so exact-name searches never lose to semantic neighbours.

8. Data Product Detail Specification

Eight tabs. The header and right rail persist across all of them so the consumer can act from any tab.

8.1 Persistent Header

- Product name, version, certification mark, sensitivity tier, deprecation banner if applicable.
- Owner block: named individual, team, escalation contact, on-call rotation link, and time since the owner last reviewed the listing.
- Five headline metrics: quality composite, freshness against SLA, active consumers (90d), monthly queries, and contract conformance percentage.
- Right rail actions: Request Access · Try an Agent · Request Enhancement · Subscribe to Changes · Copy Endpoint · Add to Comparison · Export Listing (PDF).

8.2 Tab 1 — Overview

- **Purpose and business context** — what decisions this product supports, written for a business reader.
- **Certified KPIs served** — each with its single authoritative definition, calculation expression, grain, owner and last review date. One definition per KPI is enforced: a second definition for the same KPI name is a blocking publish error.
- **Grain, coverage and history** — declared grain, entity coverage, history depth, retention policy, and known exclusions stated explicitly.
- **Update cadence** — schedule, dependency chain, and typical completion time with variance.
- **Attached agents** — cards for every agent consuming this product, each with its capability statement and a direct link to its demo.
- **Known limitations** — a mandatory, owner-authored field. Publishing is blocked while it is empty; "none" is not an accepted value.

8.3 Tab 2 — Schema & Semantics

- Column table: physical name, business name, type, nullability, classification tags, description, source lineage reference, and the DQ rules attached to that column.
- Classification tags are inherited from source and rendered read-only in this view; they are changed at source and propagate mechanically.
- Semantic layer view: dimensions, measures, hierarchies, joins and synonyms as exposed to natural-language querying.
- Sample values render only for Public and Internal tiers and only where the column carries no sensitivity tag; otherwise a masked pattern with an explanatory tooltip is displayed.
- Schema evolution timeline showing every version, the change type (additive, breaking, deprecating), the date, and the consumer notification record.

8.4 Tab 3 — Data Quality

The composite quality score is computed from six dimensions. Weights and thresholds are rubric data, versioned, and overridable per archetype — a document corpus is not scored on referential integrity.

Dimension	Default weight	Measurement
Completeness	20%	Null and default-value rates on required fields versus contract thresholds
Accuracy	20%	Conformance to reference data, validation rules and reconciliation against authoritative counts
Freshness	20%	Actual load completion versus contract SLA; measured per partition, not per table
Consistency	15%	Cross-product agreement on shared entities and KPI values; referential integrity
Validity	15%	Domain, range, format and business-rule conformance
Uniqueness	10%	Duplicate rate at the declared grain

The tab renders: the composite with a 90-day trend line; a per-dimension breakdown with the rules contributing to each; the full rule inventory with pass rates; open and recently closed incidents with MTTR; and an SLA conformance strip. Every rule is stored in the rule table with its threshold, severity and owner — no thresholds in pipeline code. Grade bands: ≥90 Exemplary, 75–89 Healthy, 60–74 Watch, 40–59 At Risk, <40 Not Fit for Consumption. Below 40 the product is automatically flagged in the catalog and agents attached to it display a trust warning.

8.5 Tab 4 — Data Contract

The contract is the enforceable promise the product makes to consumers. It is authored as YAML, versioned, and rendered in this tab both as a readable summary and as raw source.

```yaml
contract:
  product_id: DP-TEL-001
  version: 3.2.0
  owner: { name: "R. Mehta", team: "Telecom Customer Domain", escalation: "#dp-telecom" }
  grain: "one row per subscriber per day"
  schema_stability: additive_only          # breaking changes require major version + 60d notice
  guarantees:
    freshness:     { target: "06:00 local", p95_minutes: 45, measured: per_partition }
    availability:  { target_pct: 99.5, window: monthly }
    completeness:  { required_fields_pct: 99.9 }
    accuracy:      { reconciliation_variance_pct: 0.5, against: "billing_system_of_record" }
  deprecation_policy: { notice_days: 90, minimum_parallel_run_days: 30 }
  support: { hours: "24x5", p1_response_minutes: 30, on_call: "pagerduty://telecom-dp" }
  classification: { max_sensitivity: confidential, contains_pii: true, residency: ["US","EU"] }
  consumer_obligations:
    - "No redistribution outside the granted purpose"
    - "Re-derived KPIs must cite the certified definition ID"
  breach_process: "Auto-incident + consumer notification within 30 minutes of detection"
```

Rendered alongside the source: current conformance against each guarantee over 90 days, the breach history with root causes, the consumer notification log, and a diff view against any prior contract version.

8.6 Tab 5 — Endpoints & Consumption

Specified in full in Section 9. The tab presents a copy-ready connection block per surface, the authentication method, rate limits, an example call, and a live "test my access" button that reports exactly which entitlement is missing when a call would fail.

8.7 Tab 6 — Lineage & Mesh

- Upstream lineage to source systems, rendered at table level with column-level drill-down where the platform supports it.
- Downstream consumers: products, agents, reports and models that depend on this product, with dependency criticality.
- Mesh neighbourhood: sibling products sharing upstream sources, with the shared source set named and the overlap strength shown. Full mesh semantics in Section 16.
- Impact analysis: "if this product breaks, what breaks" — the transitive downstream set with consumer counts, used to justify severity at incident time.

8.8 Tab 7 — Consumption Patterns

This tab answers the owner's question — is anyone using this, and how — and the consumer's question — is this load-bearing for people like me.

Panel	Content
Adoption curve	Distinct consumers and query volume over 13 months, with certification and major-version markers annotated on the timeline
Consumer mix	Breakdown by department, role and access tier; new versus returning consumers per month
Surface mix	Query share by SQL, REST, MCP, BI tool, agent and notebook — the key signal for where to invest next
Query shape	Top 20 query patterns by frequency, most-selected columns, most-filtered columns, and columns never queried in 180 days (retirement candidates)
Peak profile	Hour-of-day and day-of-week heatmap with concurrency peaks, used for capacity planning
Cost attribution	Compute and storage cost by consumer department, with cost per query trend
Agent consumption	Share of access originating from agents versus humans, and which agents dominate
Health of consumption	Failed query rate, permission-denied rate (a leading indicator of entitlement gaps), and abandoned-session rate

8.9 Tab 8 — Value Case

Every product carries a value case reviewed at least every two quarters. Structure defined in Section 18. The tab shows the declared business outcomes, the quantified benefit model with its assumptions stated openly, realized-versus-projected tracking, the cost of ownership, and net value with a confidence rating.

9. Data Contracts & Consumption Endpoints

Five consumption surfaces, generated from one product manifest. Endpoint definitions are renderings — editing them by hand is a build error.

9.1 Surface Matrix

Surface	Consumer	Auth	Notes
SQL / semantic view	Analysts, BI tools	Platform RBAC via granted role	A governed view or Snowflake semantic view; never direct base-table grants
REST API	Applications, integrations	OAuth 2.0 client credentials, scoped to product + purpose	Cursor-paginated, filterable, JSON; 5xx and rate-limit semantics documented
MCP endpoint	AI agents, copilots, IDE assistants	OAuth 2.0 with agent identity, entitlement-scoped	Model Context Protocol server exposing typed tools per product — the primary AI consumption path
Streaming	Real-time consumers	Kafka SASL/OAuth or platform stream grants	Only for products declaring real-time or near-real-time cadence
File / share	Bulk and cross-org consumers	Signed URL or native platform share	Snowflake secure share, Delta Sharing, or object-store manifest

9.2 MCP Endpoint Specification

The MCP surface is what makes a data product consumable by any compliant agent — internal or third-party — without bespoke integration. Each product publishes a server exposing a small, typed tool set generated from its contract.

```
{
  "server": "mcp://marketplace.${TENANT}.internal/dp/DP-TEL-001",
  "product": { "id": "DP-TEL-001", "version": "3.2.0",
               "sensitivity": "confidential", "contains_pii": true },
  "auth": { "type": "oauth2", "flow": "client_credentials",
            "scopes": ["dp:DP-TEL-001:read"], "identity": "agent" },
  "tools": [
    { "name": "query_subscriber_churn",
      "description": "Return churn metrics by segment, region and period.",
      "input_schema": { "period": "string", "segment": "string?",
                        "region": "string?", "grain": "enum[day,week,month]" },
      "returns": "rows + certified KPI definition IDs + freshness_as_of",
      "row_limit": 10000, "cost_class": "small" },
    { "name": "describe_schema",  "description": "Columns, types, classifications." },
    { "name": "get_kpi_definition", "description": "Authoritative definition by KPI ID." },
    { "name": "get_freshness",     "description": "Load state vs contract SLA." },
    { "name": "get_quality",       "description": "Current composite and dimension scores." }
  ],
  "policy": {
    "row_filters_applied": true,
    "column_masking_applied": true,
    "purpose_required": true,
    "max_rows_per_call": 10000,
    "audit": "every call logged with agent_id, user_on_behalf_of, purpose, columns_returned"
  }
}
```

- **Policy is enforced server-side, never by the calling agent.** Masking and row filters are applied in the data platform; the MCP server cannot return data the caller's entitlement does not permit.
- **Every response carries provenance** — product ID, contract version, freshness timestamp, quality composite, and the certified KPI definition IDs used. This is what allows an agent to cite its sources.
- **Purpose binding is mandatory** for Confidential and Restricted products: the calling agent must pass the purpose recorded at grant time, and it is logged with the call.
- **Delegated identity is preserved.** When an agent acts for a user, both identities are carried and logged; the effective entitlement is the intersection of the two.

9.3 REST Contract Conventions

```
GET /api/v1/products/DP-TEL-001/data?period=2026-Q2&segment=postpaid&limit=1000
Authorization: Bearer <token>   X-Purpose: churn-analysis-emea   X-Request-Id: <uuid>

200 OK
{ "product": "DP-TEL-001", "contract_version": "3.2.0",
  "freshness_as_of": "2026-09-03T06:12:04Z", "quality_composite": 94,
  "kpi_definitions": ["KPI-CHURN-001","KPI-ARPU-004"],
  "rows": [ ... ], "next_cursor": "eyJvZmZzZXQiOjEwMDB9" }

403  { "error": "entitlement_missing", "required_scope": "dp:DP-TEL-001:read",
       "request_access_url": "https://marketplace/.../requests/new/access?asset=DP-TEL-001" }
```

A 403 always returns the exact missing scope and a deep link to the pre-filled access request. Dead ends are a design defect.

10. AI Agent Catalog — Browse & Discovery

The agent catalog is organized around the question the consumer wants answered, not around the technology the agent is built on. The dominant browse axis is therefore KPI coverage.

10.1 Facets

Facet	Values / behaviour
Industry / domain	Same taxonomies as data products
KPI answered	Type-ahead over the certified KPI register — "show me every agent that can answer churn rate"
Data product consumed	Filter agents by the products they run on; reachable directly from any product page
Agent archetype	Analytical Q&A · Diagnostic / root-cause · Monitoring & alerting · Forecasting · Document / RAG · Workflow-actioning
Autonomy level	L0 read-only answers · L1 answers with recommendations · L2 drafts actions for approval · L3 executes scoped actions with audit
Answer confidence	Rolling 30-day answer-acceptance rate band
Access status	Available to me · Request required · Demo only
Certification	Certified · Published · Beta · Deprecated

10.2 Agent Card Anatomy

Zone	Elements
Header	Agent name · glyph avatar · certification mark · autonomy level badge (L0–L3)
Capability statement	One sentence beginning "Answers questions about..." — a required, owner-authored field of 90–140 characters
KPI chips	Up to four certified KPIs the agent can answer, with an overflow counter
Data product strip	Icons and names of the products it consumes, each linking to that product
Metric row	Questions answered (30d) · answer acceptance rate · median latency · analyst hours deflected (30d)
Footer	Owner and team · "Try 5 questions" primary CTA · Request access secondary

The "Try 5 questions" CTA is the single most important control in the catalog. It opens the demo console pre-loaded with the agent’s curated question set and requires no access grant.

11. Agent Detail Specification & KPI Coverage Map

11.1 Tab 1 — Overview

- **What it does** — the capability statement expanded into three to five sentences of business-facing description: which decisions it supports, for which roles, at which cadence.
- **How it helps** — a required "business analysis value" block naming the analyses it performs (trend explanation, variance decomposition, cohort comparison, driver ranking, anomaly triage, scenario framing) and the manual work each replaces.
- **What it cannot do** — an explicit out-of-scope list. Stating the boundary is a trust feature, not a disclaimer; publishing is blocked while this field is empty.
- **Who it is for** — target personas and the seniority of the decisions it supports.
- **Owner block** — owner, team, on-call, model and version, last evaluation date.

11.2 Tab 2 — Capabilities & KPI Coverage Map

The coverage map is the agent's functional contract. It is a matrix, not prose, and it is generated from the agent manifest so it cannot drift from what the agent actually exposes.

Column	Meaning
KPI / question class	The certified KPI or analytical question type the agent covers
Definition ID	Link to the single authoritative definition used — the agent may not compute a KPI it does not cite
Source data product	Which product supplies the measure, with the column set used
Supported grain	Time grains and dimensional grains available for that KPI
Supported slices	Dimensions the agent can break the KPI down by
Analysis depth	Reports value · compares periods · explains variance · ranks drivers · forecasts
Confidence	Rolling accuracy for this KPI class from the evaluation suite, with sample size

Below the matrix: the example question bank grouped by KPI, the natural-language synonyms the agent recognizes, and a coverage-gap panel naming questions consumers have asked that the agent could not answer — which feeds directly into the enhancement backlog.

11.3 Tab 3 — Live Demo

Specified in Section 12.

11.4 Tab 4 — Data Products

- Every product consumed, with the specific columns and tools accessed, and whether access is read-only.
- Each product's current quality score and freshness, so a consumer can see that an agent is only as trustworthy as its inputs. If any upstream product falls below the Watch band, a banner appears on the agent detail page and on its catalog card.
- The MCP tool bindings actually granted to this agent, with the scope string for each.
- Refresh dependency: how recent the agent's answers can be, derived from the slowest upstream contract.

11.5 Tab 5 — Safety, Scope & Governance

Control	Specification
Identity	The agent holds a distinct machine identity, never a shared or human credential. All access is attributable to that identity plus the delegated user.
Entitlement scope	Explicit list of granted scopes. Effective access is the intersection of the agent scope and the invoking user's entitlements — an agent can never widen a user's access.
Data boundaries	Named products and columns the agent may reach; classified columns require an explicit exception with privacy approval recorded.
Compositional check	Where an agent composes multiple tools, the reconciliation check evaluates whether the composed result discloses more than any single tool permits; flagged combinations require review before publish.
Guardrails	Prompt-injection defences, output filters for sensitive patterns, refusal policy, and the escalation path when a question falls outside scope.
Grounding policy	Every numeric answer must cite the data product, KPI definition ID and as-of timestamp. Uncited numeric output is a blocking evaluation failure.
Human-in-the-loop	For L2 and L3 agents, which actions require approval, who approves, and where the approval is recorded.
Audit	Every question, retrieved context, tool call, returned column set, answer and user feedback is logged and replayable for the retention period.
Evaluation	Golden question set with expected answers, run on every model or prompt change; results published to this tab with pass rate and regression history.

11.6 Tab 6 — Observability

Agent-specific health: question volume and trend, answer acceptance and thumbs-down rate, groundedness (share of numeric claims with valid citations), refusal and fallback rates, p50/p95 latency, cost per answer, token consumption, upstream product health, and evaluation-suite drift. Detail in Section 19.

11.7 Tab 7 — Value & Adoption

Distinct users and repeat-use rate, questions answered, estimated analyst hours deflected with the derivation shown, decisions influenced where captured, cost per answer versus the analyst-hour baseline, and the adoption curve with annotations for capability releases. Method in Section 18.

12. Live Demo Console — Guided Question & Answer

The demo console is the conversion surface of the entire product. Every agent must ship with at least five curated exchanges before it can be published; the publish gate checks the count and rejects below five.

12.1 Layout

- **Left panel (320px):** the curated question list. Each entry shows the question, the KPI class it exercises, and the analysis type. Clicking runs it.
- **Centre stage:** conversation thread. Question renders as typed input, then the answer streams token by token exactly as in production.
- **Answer composition:** a one-sentence headline finding, a supporting visualization, a compact data table, the KPI definition used, an explicit as-of timestamp, and a citation block naming the data products and columns consulted.
- **Right rail:** "How this answer was produced" — the tool calls made, the products queried, the rows scanned, latency and cost. Transparency is the demo's persuasive core, not a debug view.
- **Below the thread:** free-text input labelled with the active tier (Demo data / Live data), plus follow-up chips suggesting the next question.

12.2 Curated Exchange Definition

Each of the five-plus exchanges is authored as data and stored with the agent manifest:

```
demo_exchanges:
  - id: DEMO-AG-TEL-001-03
    question: "Which customer segments drove the churn increase last quarter?"
    kpi_class: "KPI-CHURN-001"
    analysis_type: driver_ranking
    expected_answer_shape:
      headline: "single sentence naming the top driver segment and its contribution"
      visual: bar_chart          # segment on x, churn delta on y
      table_columns: [segment, churn_rate_prior, churn_rate_current, delta_pp, contribution_pct]
      must_cite: ["DP-TEL-001", "KPI-CHURN-001"]
    data_tier: demo              # demo | live
    max_latency_ms: 6000
    evaluation:
      golden_answer_ref: "eval/AG-TEL-001/q3.json"
      tolerance_pct: 2.0
    last_validated: 2026-08-28
```

12.3 Rules of the Console

Rule	Rationale
No mocked answers	The console executes the real agent against a real dataset — demo tier uses a synthetic dataset generated to be statistically realistic and referentially complete. A recorded or hardcoded answer is prohibited.
Tier is always visible	A persistent badge states Demo data or Live data. Demo-tier answers carry a footer line stating the data is synthetic.
Demo tier is never real regulated	Restricted-tier products never render real values in the console under any role.

Rule	Rationale
data	
Every numeric claim is cited	Uncited numbers do not render; the answer is returned as a failure with an explanation instead.
Curated exchanges are validated nightly	A scheduled job runs all curated questions against the golden answers. A failure marks the exchange stale, removes it from the front-page theatre, and notifies the owner.
Latency budget	p95 under 6 seconds for curated questions. Slower agents are excluded from the front-page theatre until remediated.
Feedback is captured	Thumbs up/down plus an optional reason on every answer, written to the evaluation corpus and surfaced in the owner console.

12.4 Demo Data Tier

- A synthetic dataset per data product, generated to preserve distributions, seasonality, referential integrity and cardinality without containing any real subject data.
- Generated from the product contract and profile statistics; regenerated whenever the schema version changes so the demo never drifts from the live shape.
- Explicitly labelled at every touchpoint and stored in a physically separate schema with no path to production data.
- Sized for sub-second query response so demo latency is dominated by model inference rather than scan time.

13. Access Request & Entitlement Workflow

One workflow engine serves all three request types. Access requests are the highest-volume path and the one whose latency most directly determines whether the marketplace succeeds.

13.1 Request Form

Field	Behaviour
Assets	Pre-filled from context. Multi-asset requests are supported and evaluated as one unit so a consumer never files four tickets for one analysis.
Access level	Read metadata · Read data · Read data with PII · Agent invocation · Write-back (where the product supports it)
Purpose	Required free text plus a controlled purpose category. The purpose is bound to the grant, logged with every query, and re-verified at renewal.
Duration	30 / 90 / 180 / 365 days or project-bound. Perpetual access requires administrator justification and is reviewed quarterly.
Consumption surface	Which endpoints are needed — determines which scopes are provisioned
Legal basis / attestation	Required for Confidential and Restricted tiers; renders the specific obligations from the contract and requires explicit acceptance
Cost centre	Required where the product charges back; drives cost attribution in Section 18
Downstream sharing	Declaration of whether outputs will be shared beyond the requester's team, and with whom

13.2 Policy Evaluation — Runs Before Submission

The moment the form is complete, the policy engine evaluates the request and shows the consumer exactly what will happen. No blind submissions.

8. **Auto-approve** — requester role is in the product's pre-approved list, sensitivity is Internal or below, no PII, and purpose category is permitted. Provisioned immediately; owner notified.
9. **Owner approval** — default path for Internal and Confidential assets. SLA: 2 business days.
10. **Owner + steward approval** — Confidential assets containing classified columns. SLA: 3 business days.
11. **Owner + privacy + security approval** — Restricted tier, cross-border access, or any PII-inclusive grant. SLA: 5 business days.
12. **Blocked** — the request violates a hard policy (residency, licence restriction, segregation of duties). The consumer is told which policy and offered an alternative asset where the mesh identifies a lower-sensitivity equivalent.

13.3 Approval Experience

- Approvers act from the console, from email, or from Slack/Teams — the same decision object, whichever surface is used.
- The approval card shows the requester, their organizational unit, the purpose, the assets, what the requester already has access to, and any prior denials — enough to decide in under a minute.
- Delegation and out-of-office fallback are configured per owner; an unattended request escalates to the domain steward at 80% of SLA and to the architect at breach.

- Partial approval is supported: an approver may grant a subset of assets or downgrade the access level, with a required reason that is returned to the requester.
- Every decision records actor, timestamp, decision, reason and the policy version in force.

13.4 Provisioning

On approval, provisioning is mechanical. Nobody edits a role by hand.

```
on_approval:
  1. resolve target platform role      -> role naming convention: MKT_<PRODUCT>_<LEVEL>
  2. grant scoped role to principal    -> Snowflake / UC / IAM via connector
  3. apply masking + row policy         -> inherited from product classification
  4. bind purpose + expiry to grant     -> written to entitlement register
  5. issue OAuth scopes for REST + MCP  -> dp:<id>:read, agent:<id>:invoke
  6. notify consumer                    -> runbook, endpoints, quickstart, academy module
  7. write immutable audit record       -> request, decision chain, grant, expiry

on_expiry_minus_14d:  notify consumer + owner, offer one-click renewal
on_expiry:            revoke automatically, retain audit record, log revocation
on_no_use_60d:        flag dormant grant to owner for revocation (least privilege hygiene)
```

13.5 Consumer-Side Views

- **My access** — everything I hold, with expiry countdowns, purpose, usage in the last 30 days, and one-click renewal or self-revocation.
- **My requests** — status, current approver, SLA clock, and history.
- **Approvals queue** — for approvers, with ageing, bulk action for like-for-like requests, and a breach indicator.
- **Entitlement register** — for security officers: every grant across the estate, filterable by product, person, purpose, sensitivity and expiry, exportable for audit.

14. Enhancement Request Workflow

Enhancement requests are how a published asset improves. They are filed against an existing product or agent and land in that owner's public backlog.

14.1 Enhancement Types

Type	Examples	Typical path
New attribute / column	Add customer tenure band; add region hierarchy level	Owner assessment → source feasibility → schema version
New KPI or measure	Add net revenue retention to the product's certified KPI set	Definition review by steward → semantic layer change → agent coverage update
Grain or history change	Move from weekly to daily; extend history from 13 to 36 months	Cost and capacity assessment → contract amendment
Freshness improvement	Move from daily 06:00 to hourly	Pipeline redesign → contract SLA amendment → consumer notification
Quality rule addition	Add validity rule on a code field that is silently drifting	Rule authored as data → threshold agreed → monitored
New endpoint / surface	Expose an MCP tool for a currently SQL-only product	Manifest change → generated rendering → scope definition
Agent capability	Teach the agent to answer a KPI it currently refuses	Coverage-gap validation → tool binding → evaluation set extension
Documentation / usability	Business names missing on 40 columns; unclear limitation statement	Steward assignment → fast track

14.2 Lifecycle

Submitted → Triaged → Assessed → Accepted / Declined / Merged → Scheduled → In progress → Delivered → Verified. Every transition is timestamped and visible to the requester and to anyone watching the asset.

- **Triage SLA: 5 business days.** The owner must acknowledge with a disposition; silence escalates to the domain steward.
- **Assessment** records effort estimate, impact estimate, affected consumers, and whether the change is additive or breaking.
- **Decline requires a reason** drawn from a controlled list (out of scope, source unavailable, cost prohibitive, duplicate, superseded, security constraint) plus free text. Declines are public.
- **Merge** links duplicate requests into one canonical item and carries all requesters' votes and context forward.
- **Delivery** links the enhancement to the contract version that shipped it and notifies every subscriber and requester.
- **Verification** asks the original requester to confirm the enhancement met the need; unverified items after 14 days are auto-closed as delivered with a note.

14.3 Public Backlog

Each product and agent detail page carries a backlog panel showing open enhancements with vote counts, status and target release. Consumers can upvote and subscribe. Visible demand is what allows an owner to defend investment — and what stops the same request arriving five times from five teams.

15. Demand Intake — New Data Product & Agent Requests

When the catalog does not serve the need, the need must not vanish. Demand intake captures it as a first-class object with the same lifecycle rigour as published supply.

15.1 Intake Form

Section	Fields
What	Request type (data product / agent / both) · working title · the business question that cannot be answered today
Why	Business outcome · decision it enables · quantified value if known · consequence of not building it
Who	Requesting team · expected consumer population · executive sponsor · other teams likely to benefit
Data	Known source systems · required entities and attributes · required grain · required history · required freshness
Agent (if applicable)	Questions it must answer · KPIs involved · required autonomy level · target personas
Constraints	Regulatory considerations · residency · deadline and its driver · budget or cost centre
Evidence	Existing workarounds — spreadsheets, extracts, shadow pipelines — that this would replace

15.2 Duplicate-Supply Detection

At submission the intake engine runs a semantic match of the request against the published catalog, the demand board, and in-flight roadmap items. This is the highest-leverage control in the entire product: it converts a build request into a reuse decision.

```
similarity = 0.40 · embedding_match(request_text, product.purpose + kpi_definitions)
           + 0.25 · entity_overlap(request.entities, product.entities)
           + 0.20 · source_overlap(request.sources, product.upstream_sources)
           + 0.15 · kpi_overlap(request.kpis, product.certified_kpis)

similarity >= 0.75  -> "This may already exist"  : blocking review with the owner before
                        the request can progress; requester shown the candidate side by side
0.50 - 0.74         -> "Related supply exists"   : advisory panel, requester may proceed
< 0.50              -> proceed to scoring

Every match stores { candidate_id, similarity, contributing_factors, confidence, rationale }.
Matches below 0.80 confidence are reviewed by an architect before being shown as blocking.
```

15.3 Demand Scoring

Requests are scored on a rubric stored as data. The score orders the demand board but never decides alone — the architect triage panel makes the call and records a reason.

Criterion	Weight	Scale
Business value	30%	Quantified benefit, or a qualitative band with sponsor attestation
Consumer breadth	20%	Distinct teams and estimated consumers; votes on the demand board
Strategic alignment	15%	Mapping to declared enterprise priorities for the period

Criterion	Weight	Scale
Feasibility	15%	Source availability, quality of source, engineering effort
Reuse leverage	10%	Extent to which the new product would serve future requests or unblock agents
Risk / compliance urgency	10%	Regulatory driver or risk-reduction value

15.4 Demand Board

- Public, filterable board of all open demand with status, score, vote count, requesting teams and target quarter.
- Voting and "me too" with a required one-line use case — votes without context are not counted, because the use case is what makes clustering possible.
- Automatic clustering of semantically similar requests into a demand theme; a theme with five or more distinct requesting teams is auto-escalated to the portfolio review.
- Status lifecycle: Submitted → Duplicate review → Triaged → Scored → Roadmapped → In build → Delivered, or Declined with a public reason.
- On delivery, the demand item links to the published listing and every requester and voter is notified — closing the loop is what sustains submission volume.

15.5 Intake to Blueprint

An accepted demand item generates a draft product or agent manifest pre-filled from the request: proposed name, domain, archetype, entity list, candidate sources drawn from lineage, draft contract skeleton, and — for agents — a draft KPI coverage map and five candidate demo questions derived from the questions in the request. The draft carries confidence and rationale on every inferred field and requires human completion and approval before it can be built. The specification is never auto-approved.

16. Data Product Mesh

The mesh makes the shape of the estate visible: which products are related, which share upstream sources, where duplication is forming, and what breaks when a source fails. Edges are computed from lineage and metadata; none are drawn by hand.

16.1 Edge Types

Edge	Derivation	Meaning to the consumer
Shared source	Jaccard overlap of upstream source tables from platform lineage	These products describe the same underlying reality and should agree
Direct dependency	One product materially consumes another	A hard dependency — upstream incidents propagate
Shared entity	Both products key on the same business entity and identifier	Joinable at a known key; candidate for a combined analysis
Shared KPI	Both expose the same certified KPI definition	Consistency requirement — divergent values are an incident
Semantic similarity	Embedding similarity of purpose, columns and glossary terms	Possible duplication; carries confidence and rationale
Co-consumption	The same consumers query both within a session window	Behavioural affinity — drives "used together" recommendations

16.2 Edge Weighting

```
edge_strength = 0.30 · source_overlap_jaccard
              + 0.25 · entity_overlap
              + 0.20 · kpi_overlap
              + 0.15 · semantic_similarity
              + 0.10 · co_consumption_lift

render_threshold: 0.25          # weights and threshold are rubric data, tunable per tenant
duplication_alert: >= 0.80 on (source_overlap AND semantic_similarity)
every edge stores: { type, strength, contributing_factors, confidence, rationale, computed_at }
```

16.3 Explorer Interactions

- **Layouts:** force-directed by default; also domain-clustered, industry-clustered, and source-anchored (sources as hub nodes).
- **Node encoding:** size = active consumers; colour = quality band; ring = certification; pulse = open incident.
- **Hover** reveals the neighbourhood with edge types and the named shared sources.
- **Blast-radius mode:** select a source system and highlight every product and agent downstream, with consumer counts — the fastest incident-severity assessment in the product.
- **Duplication mode:** show only high-similarity edges, listing candidate consolidations with estimated savings.
- **Gap mode:** overlay demand-board themes onto the mesh to show where the estate has no supply.
- **Accessible equivalent:** every mesh view has a keyboard-navigable table rendering the same edges, counts and attributes. The graph is never the only path to the information.

17. AI Agent Mesh

The agent mesh answers three questions: which agents do similar things, which agents depend on the same data, and where should a consumer be routed when one agent cannot help.

17.1 Edge Types

Edge	Derivation	Use
Shared data product	Both agents consume the same product	Upstream incident impact; consolidation candidate
Overlapping KPI coverage	Coverage maps intersect on certified KPI IDs	Duplication risk; a KPI answered by two agents must not diverge
Complementary	One agent’s out-of-scope questions are another’s coverage	Powers "this agent cannot answer that — try this one" handoff
Compositional	One agent invokes another as a tool	Explicit dependency; triggers the compositional entitlement check
Same domain	Shared industry and business domain	Clustering for browse
Co-usage	Same users query both in a session window	Behavioural affinity for recommendations

17.2 Governance Signals Surfaced by the Agent Mesh

- **Divergence detection.** Where two agents answer the same certified KPI, a scheduled job asks both the same question and compares. A material difference is an incident against both, because a KPI with two answers is worse than a KPI with none.
- **Consolidation candidates.** Agents with coverage overlap above 0.70 and shared products above 0.70 are surfaced to the domain architect with a recommendation and a rationale.
- **Coverage gap map.** Overlay of consumer questions that no agent answered onto the domain grid — the direct input to the agent roadmap.
- **Compositional exposure.** Where agent A invokes agent B, the mesh renders the union of data reachable through the chain, and flags where that union exceeds what either agent is individually scoped for. This check runs at publish time and on every scope change.

17.3 Handoff Behaviour

When an agent receives a question outside its coverage map it does not guess. It states the boundary, names the agent or data product that does cover it, and offers a one-click handoff carrying the question across. Where nothing covers it, the refusal offers a pre-filled demand request. Every out-of-scope question is logged to the coverage-gap corpus regardless of what the consumer does next.

18. Usage, Adoption & Value Realization

Adoption without value is vanity; value without adoption is a projection. The module tracks both and shows the arithmetic openly — an unfalsifiable value claim is worse than no claim.

18.1 Metric Layers

Layer	Metrics
Usage	Queries, API calls, MCP tool invocations, agent questions, distinct users, sessions, rows scanned, cost consumed
Adoption	Distinct consumers (30/90d), new versus returning, team penetration versus addressable population, repeat-use rate, time-to-first-query after grant, breadth across departments
Engagement quality	Session depth, follow-up question rate, answer acceptance rate, abandonment rate, permission-denied rate
Value	Analyst hours deflected, duplicate builds avoided, decision cycle-time reduction, revenue or cost outcomes attributed by the business, risk events prevented
Cost	Compute, storage, model inference, platform licence share, and stewardship effort — allocated per asset and per consuming department
Net	Value minus cost, with a confidence rating and the assumption set displayed

18.2 Value Case Structure

Every published asset carries this structure. It is authored by the owner, reviewed by the sponsor, and re-reviewed at least every two quarters. Staleness is displayed on the listing.

Component	Requirement
Business outcome	The named decision or process this asset improves — not "better insights"
Baseline	How the outcome was achieved before, with time and cost, captured at intake
Benefit model	Explicit formula with named inputs, e.g. deflected_questions × avg_analyst_minutes × loaded_rate
Assumptions	Every assumption stated and dated. Unstated assumptions make the case indefensible at budget review.
Realized value	Measured to date from telemetry, versus the projection
Attribution confidence	High (measured directly) · Medium (modelled from telemetry) · Low (estimated). Displayed with the number, always.
Cost of ownership	Build, run, and stewardship cost
Review record	Last reviewed date, reviewer, and change from prior review

18.3 Deflection Model for Agents

```
deflected_hours =  answered_questions
                 × answer_acceptance_rate
                 × avg_manual_minutes_per_question_class   -- calibrated per KPI class
                 ÷ 60

deflected_value =  deflected_hours × loaded_analyst_rate
net_value       =  deflected_value - (inference_cost + platform_cost + stewardship_cost)

Calibration: avg_manual_minutes is sampled quarterly from an analyst time study per
question class and stored as reference data with the sample size and date. It is never
a hardcoded constant, and the sample size is displayed next to any figure derived from it.
```

18.4 Portfolio Views

- **Value versus cost quadrant** across all assets — high value / low cost is the reference pattern; low value / high cost is the retirement queue.
- **Adoption cohort curves** by publication quarter, showing whether newer assets reach adoption faster than older ones.
- **Retirement candidates** — assets with declining consumers, no queries in 90 days, or unresolved quality decline, with a structured deprecation proposal.
- **Unmet demand value** — the aggregate scored value sitting on the demand board, which is the argument for platform headcount.
- **Executive board pack** — a generated PDF/PPTX from the same snapshot data, so the deck and the dashboard can never disagree.

19. Observability for Data Products & Agents

One observability plane covering both asset classes, because an agent's health is inseparable from the health of the products beneath it.

19.1 Data Product Signals

Signal group	Measures	Alert condition
Freshness	Load completion time per partition versus contract SLA; lag distribution	SLA breach, or p95 lag trending beyond 80% of budget
Volume	Row counts versus forecast band; partition completeness	Deviation beyond the seasonally-adjusted band
Schema	Column additions, type changes, drops; contract conformance	Any breaking change, or any change without a version bump
Quality	Rule pass rates by dimension; composite score movement	Composite drop of more than 5 points, or any critical rule failure
Distribution	Null rates, cardinality, categorical mix, numeric moments versus baseline	Drift beyond configured tolerance — a leading indicator of silent upstream change
Pipeline	Run duration, failure and retry rates, upstream dependency state	Failure, or duration beyond p95 baseline
Access	Query success rate, permission-denied rate, unusual access patterns	Denied-rate spike (entitlement gap) or anomalous access (security)
Cost	Compute and storage trend per consumer	Cost per query rising while query volume is flat

19.2 Agent Signals

Signal group	Measures	Alert condition
Answer quality	Acceptance rate, thumbs-down rate, evaluation-suite pass rate	Pass rate below threshold or a 3-point week-over-week acceptance drop
Groundedness	Share of numeric claims carrying valid citations; citation validity checks	Any uncited numeric output — treated as a defect, not a warning
Coverage	In-scope rate, refusal rate, out-of-scope question volume by theme	Refusal rate above threshold, or a new gap theme reaching volume
Performance	p50/p95 latency, timeout rate, tool-call failure rate	p95 beyond budget; any tool-call error-rate spike
Cost	Tokens and inference cost per answer; cost per accepted answer	Cost per accepted answer rising faster than volume
Upstream health	Quality and freshness of every consumed product	Any upstream product entering the Watch band — surfaces a trust banner on the agent
Drift	Evaluation-suite regression across model or prompt versions	Any regression against the golden set blocks promotion
Safety	Guardrail triggers, injection attempts, scope-violation attempts	Any scope-violation attempt is a security event,

Signal group	Measures	Alert condition
		logged and routed

19.3 Incident Lifecycle

Detect → classify severity by blast radius from the mesh → notify owner and affected consumers → display a banner on every affected listing → track to resolution → publish a root cause → link the incident permanently to the asset's quality history.

- Severity is computed, not chosen: it derives from downstream consumer count, sensitivity tier, and contract guarantee breached.
- Consumer notification is automatic for any breach of a contract guarantee. Owners cannot suppress consumer notification; they can only add context.
- Every incident appears on the product's quality tab and in the agent trust banner for as long as it is open, and in the history permanently.
- MTTR by product, by domain and by owner is a standing metric on the executive dashboard.

20. Marketplace Academy

The academy exists so that adoption is not gated on tribal knowledge. It is embedded in the product rather than parked on a separate learning platform, and it is contextual — the right module is one click from the listing that raised the question.

20.1 Learning Paths

Path	Audience	Modules	Outcome
Consuming data products	Business consumers, analysts	Finding the right product · reading a quality score · understanding a data contract · requesting access · your first query · citing certified KPIs	Certified Consumer
Working with agents	All users	What agents can and cannot do · asking answerable questions · reading citations and as-of stamps · when to escalate to a human · giving useful feedback	Certified Agent User
Owning a data product	Product owners	Writing a contract · designing quality rules as data · defining certified KPIs · publishing and versioning · running the backlog · defending the value case	Certified Product Owner
Building an agent	Agent owners, engineers	Coverage-map design · tool scoping and least privilege · grounding and citation discipline · curating demo exchanges · building the evaluation set · publishing safely	Certified Agent Owner
Governing the estate	Stewards, architects	Classification propagation · certification workflow · mesh interpretation · duplicate-supply triage · entitlement review	Certified Steward
Executive briefing	Sponsors, leadership	Reading the portfolio view · interpreting value cases and their confidence · what to ask at portfolio review	Completion badge

20.2 Delivery Mechanics

- **Contextual runbooks** — every product and agent listing links to the exact module for consuming that asset class and surface, with the asset's own endpoints substituted into the examples.
- **Interactive sandboxes** — modules run against the demo tier, so a learner writes a real query or asks a real agent question inside the lesson.
- **Assessment and certification** — short assessments per module; certification unlocks pre-approved access tiers for that asset class, which turns learning into a concrete access benefit.
- **Glossary integration** — every certified KPI and business term has an academy card explaining it in plain language, reachable from any place the term appears.
- **Owner enablement** — templates for contracts, quality rule sets, coverage maps, demo exchange sets and value cases, all pre-populated with worked examples from the seed catalog.
- **Progress and analytics** — per-user progress, per-team completion, and correlation of certification with adoption, which is how the academy earns its own budget.

21. Canonical Data Model & Physical Schema

The canonical model is the contract between every module in the product. Modules exchange these entities and nothing else; a module that needs a new attribute extends the canonical model rather than passing an untyped payload.

21.1 Entity Groups

Group	Entities
Reference	tenant · industry · business_domain · product_archetype · sensitivity_tier · purpose_category · source_system
Identity	party (person, team, service, agent identity) · org_unit · role_assignment
Supply — data	data_product · data_product_version · data_product_column · data_contract_version · endpoint · kpi_definition · glossary_term
Supply — agents	agent · agent_version · agent_kpi_coverage · agent_product_binding · agent_tool_binding · demo_exchange · evaluation_case
Quality	quality_rule · quality_result · quality_score_snapshot · incident · incident_impact
Lineage & mesh	lineage_edge · mesh_edge_data · mesh_edge_agent
Demand & workflow	request · request_item · approval_step · decision · demand_vote · demand_theme · enhancement
Entitlement	entitlement_grant · grant_scope · purpose_binding · revocation
Telemetry	usage_event · usage_daily_agg · agent_interaction · answer_feedback · cost_allocation
Value	value_case · value_assumption · value_measurement
Governance config	rubric · rubric_version · rubric_criterion · policy · policy_version
Academy	academy_module · learning_path · enrollment · assessment_result · certification
Audit	audit_event · publication_snapshot

21.2 Core Relationships

```
data_product 1—* data_product_version 1—* data_product_column
data_product_version 1—1 data_contract_version 1—* guarantee
data_product_version *—* kpi_definition            (product serves KPI)
data_product_version 1—* endpoint                  (sql | rest | mcp | stream | file)
data_product      *—* source_system               (via lineage_edge, transitive)
data_product      *—* data_product   [mesh_edge_data: type, strength, confidence]

agent 1—* agent_version
agent_version *—* data_product_version  [agent_product_binding: columns, tools, scope]
agent_version 1—* agent_kpi_coverage    —1 kpi_definition   (must cite a definition)
agent_version 1—* agent_tool_binding    —1 endpoint(mcp)
agent_version 1—* demo_exchange         (>= 5 enforced at publish)
agent_version 1—* evaluation_case
agent         *—* agent  [mesh_edge_agent: type, strength, confidence]

request 1—* request_item —* {data_product | agent}
request 1—* approval_step 1—1 decision
request 1—? entitlement_grant 1—* grant_scope, 1—1 purpose_binding

kpi_definition  1—1 owner(party)   -- exactly one active definition per KPI name per tenant
value_case      1—* value_assumption, 1—* value_measurement
rubric 1—* rubric_version 1—* rubric_criterion   -- all weights & thresholds live here
```

21.3 Selected Physical Schema

Illustrative DDL for the entities most likely to be misimplemented. Full DDL is generated from the canonical model definition file at build time.

```
CREATE TABLE data_product (
  product_id        VARCHAR PRIMARY KEY,        -- DP-<IND>-<NNN>
  tenant_id         VARCHAR NOT NULL,
  name              VARCHAR NOT NULL,
  purpose           VARCHAR NOT NULL,           -- rendered on the card, <= 400 chars
  industry_code     VARCHAR NOT NULL REFERENCES industry(code),
  domain_code       VARCHAR NOT NULL REFERENCES business_domain(code),
  archetype_code    VARCHAR NOT NULL,
  sensitivity_tier  VARCHAR NOT NULL,           -- derived: max over columns; not editable here
  certification     VARCHAR NOT NULL,           -- certified|published|beta|deprecated
  owner_party_id    VARCHAR NOT NULL,
  current_version   VARCHAR NOT NULL,
  known_limitations VARCHAR NOT NULL,           -- publish blocked while empty
  created_at TIMESTAMP_NTZ, updated_at TIMESTAMP_NTZ
);

CREATE TABLE quality_score_snapshot (
  snapshot_id     VARCHAR PRIMARY KEY,
  product_id      VARCHAR NOT NULL,
  rubric_version  VARCHAR NOT NULL,             -- score is meaningless without it
  composite       NUMBER(5,2) NOT NULL,
  completeness NUMBER(5,2), accuracy NUMBER(5,2), freshness NUMBER(5,2),
  consistency  NUMBER(5,2), validity NUMBER(5,2), uniqueness NUMBER(5,2),
  band            VARCHAR NOT NULL,             -- from rubric bands, not code
  evidence_ref    VARIANT NOT NULL,             -- rule_ids + results that produced it
```

```
  computed_at     TIMESTAMP_NTZ NOT NULL
);  -- immutable; never updated in place

CREATE TABLE agent_kpi_coverage (
  coverage_id     VARCHAR PRIMARY KEY,
  agent_version_id VARCHAR NOT NULL,
  kpi_id          VARCHAR NOT NULL REFERENCES kpi_definition(kpi_id),
  source_product_id VARCHAR NOT NULL,
  supported_grains  ARRAY,          -- [day, week, month, quarter]
  supported_slices  ARRAY,          -- [segment, region, channel, ...]
  analysis_depth    VARCHAR,        -- report|compare|explain|rank_drivers|forecast
  eval_accuracy     NUMBER(5,2),
  eval_sample_size  NUMBER,
  UNIQUE (agent_version_id, kpi_id)
);

CREATE TABLE mesh_edge_data (
  edge_id     VARCHAR PRIMARY KEY,
  product_a   VARCHAR NOT NULL, product_b VARCHAR NOT NULL,
  edge_type   VARCHAR NOT NULL,   -- shared_source|dependency|shared_entity|shared_kpi|
                                  -- semantic|co_consumption
  strength    NUMBER(4,3) NOT NULL,
  factors     VARIANT NOT NULL,   -- per-component contributions
  confidence  NUMBER(4,3) NOT NULL,
  rationale   VARCHAR NOT NULL,   -- human-readable; required for every edge
  reviewed_by VARCHAR,            -- required when confidence < 0.80
  computed_at TIMESTAMP_NTZ NOT NULL
);

CREATE TABLE entitlement_grant (
  grant_id     VARCHAR PRIMARY KEY,
  request_id   VARCHAR NOT NULL,
  principal_id VARCHAR NOT NULL,      -- person or agent identity
  asset_type   VARCHAR NOT NULL, asset_id VARCHAR NOT NULL,
  access_level VARCHAR NOT NULL,
  purpose_code VARCHAR NOT NULL, purpose_text VARCHAR NOT NULL,
  platform_role VARCHAR NOT NULL,     -- the role actually granted
  oauth_scopes ARRAY,
  granted_at TIMESTAMP_NTZ, expires_at TIMESTAMP_NTZ,
  revoked_at TIMESTAMP_NTZ, revocation_reason VARCHAR,
  last_used_at TIMESTAMP_NTZ          -- drives dormant-grant hygiene
);
```

21.4 Model Invariants

- Exactly one active definition per KPI name per tenant. A second is a blocking publish error, not a warning.
- A quality score row without a rubric_version is invalid and rejected at write time.
- Every mesh edge and every automated inference carries confidence and rationale; below 0.80 requires reviewed_by before display.
- An agent_version cannot be published with fewer than five demo_exchange rows, or with any agent_kpi_coverage row lacking a kpi_id.
- sensitivity_tier on a product is derived from its columns and is not directly writable.
- Publication snapshots are append-only. A published listing can be superseded, never edited.

22. Reference Architecture & Technology Stack

22.1 Layered Architecture

```
┌──────────────────────────────────────────────────────────────────────┐
│ EXPERIENCE   Next.js portal · Owner & steward consoles · Admin       │
│              Demo console · Mesh explorer · Board-pack export        │
├──────────────────────────────────────────────────────────────────────┤
│ API          REST /api/v1 · GraphQL (read) · MCP gateway · Webhooks  │
│              OAuth 2.0 / OIDC · rate limiting · purpose enforcement  │
├──────────────────────────────────────────────────────────────────────┤
│ SERVICES     Catalog · Search & Ranking · Mesh Engine · Workflow     │
│              Entitlement · Quality · Observability · Value · Academy │
│              Agent Registry · Demo Runner · Intake & Duplicate Match │
├──────────────────────────────────────────────────────────────────────┤
│ METADATA     Postgres (canonical model, rubrics, workflow, audit)    │
│              Vector index (semantic search, duplicate detection)     │
│              Object store (manifests, snapshots, exports)            │
├──────────────────────────────────────────────────────────────────────┤
│ CONNECTORS   Snowflake · Databricks · Fabric · BigQuery · Oracle     │
│              Collibra · Alation · Purview · Atlan · DataHub          │
│              Monte Carlo · Soda · Great Expectations · dbt           │
│              Okta/Entra · ServiceNow/Jira · Slack/Teams              │
├──────────────────────────────────────────────────────────────────────┤
│ DATA PLANE   Customer platform. Read-only metadata by default.       │
│              Data never leaves the customer boundary; the            │
│              marketplace stores metadata, contracts and telemetry.   │
└──────────────────────────────────────────────────────────────────────┘
```

22.2 Stack

Layer	Choice	Rationale
Frontend	Next.js App Router, TypeScript, Tailwind, shadcn/ui, Recharts, D3 for mesh	Server components keep the landing page inside its LCP budget; D3 needed for mesh interaction
API	FastAPI (Python) or NestJS (TypeScript); OpenAPI 3.1 generated from source	Contract-first; SDKs generated, never hand-written
Workflow	Temporal (or equivalent durable execution)	Approval chains with SLA timers, escalation and compensation must survive restarts
Metadata store	PostgreSQL 16 with pgvector	Relational integrity for the canonical model plus co-located embeddings for search
Search	Hybrid: pgvector or Cortex Search for semantic, Postgres FTS for lexical, fused by RRF	Exact-name search must never lose to a semantic neighbour
Agent runtime	Snowflake Cortex Agents / Snowflake Intelligence as reference; adapters for Bedrock, Vertex, Azure AI	Native runtime keeps data in the platform boundary and simplifies security review
MCP gateway	Standalone MCP server per data product, generated from the manifest	Any compliant agent can consume any product with no bespoke integration

Layer	Choice	Rationale
Observability	OpenTelemetry traces and metrics; platform-native quality signals harvested by connector	One trace spans portal → API → MCP → platform query
Deployment	Containers on Kubernetes, or Snowpark Container Services for the Snowflake-native edition	Native deployment is a structural advantage in security review and distribution
IaC & CI	Terraform, GitHub Actions; manifests and rubrics versioned in Git and promoted by pipeline	Rubric changes are reviewed like code even though they are data

22.3 Snowflake-Native Edition

- Marketplace metadata stored in a dedicated database; portal served from Snowpark Container Services so no data leaves the account.
- Semantic views back Cortex Analyst for natural-language querying of each data product; the certified KPI definitions are the semantic model's measures.
- Cortex Search services back document-corpus products and the marketplace's own search over descriptions and glossary.
- Cortex Agents implement catalogued agents; agent tool scopes map to Snowflake roles, and masking and row-access policies enforce entitlement at the data layer.
- ACCOUNT_USAGE and ORGANIZATION_USAGE supply usage, cost, access and lineage telemetry; Data Metric Functions supply quality results.
- Products may be distributed across accounts via secure shares and internal listings; the marketplace is the governance layer above those shares.

23. API Specification

Every portal capability is available through the API; the portal is a client of its own API with no privileged path. Versioned under /api/v1, OpenAPI 3.1, cursor pagination, RFC 7807 problem responses.

23.1 Endpoint Surface

Method & path	Purpose
GET /products	Search and filter the catalog; supports all facets, ranking and cursor pagination
GET /products/{id}	Full listing including contract, endpoints, quality, adoption and value
GET /products/{id}/quality	Current and historical scores with contributing rule results
GET /products/{id}/contract	Contract source and conformance history
GET /products/{id}/consumption	Usage and adoption analytics for the caller's permitted scope
GET /products/{id}/mesh	Neighbourhood with edge types, strengths, confidence and rationale
GET /products/{id}/lineage	Upstream and downstream lineage
POST /products	Create from manifest (owner role); returns validation report
POST /products/{id}/versions	Publish a new version; runs the publish gate
GET /agents	Search and filter agents, including by KPI and by data product
GET /agents/{id}	Full listing including coverage map, scope, demo set and telemetry
GET /agents/{id}/coverage	KPI coverage map with evaluation accuracy per KPI
GET /agents/{id}/demo	Curated exchanges with expected shapes and validation status
POST /agents/{id}/ask	Execute a question. Body: question, tier (demo|live), purpose. Returns answer, citations, tool trace, latency and cost
POST /agents/{id}/feedback	Record acceptance or rejection with a reason; writes to the evaluation corpus
GET /mesh/data · GET /mesh/agents	Mesh graphs with filters, thresholds and layout hints
POST /requests/access	Create an access request; returns the evaluated approval path and SLA
POST /requests/enhancement	Create an enhancement request against an asset
POST /requests/supply	Create a demand request; returns duplicate-match candidates with similarity
GET /requests · GET /requests/{id}	Status, approval chain, decisions, SLA state
POST /requests/{id}/decision	Approve, decline or partially approve with a required reason
GET /entitlements	Grants for the caller, or the full register for security roles
DELETE /entitlements/{id}	Self-revoke or administrative revoke
GET /demand	Demand board with clustering, scoring and votes

Method & path	Purpose
POST /demand/{id}/vote	Vote with a required use-case line
GET /observability/{assetType}/{id}	Signals, incidents and alerts
GET /value/{assetType}/{id}	Value case, assumptions and realized measurements
GET /academy/paths · POST /academy/enrollments	Learning paths, progress and certification
GET /admin/rubrics · POST /admin/rubrics/{id}/versions	Read and version scoring rubrics — the only way weights change

23.2 Agent Invocation Contract

```
POST /api/v1/agents/AG-TEL-001/ask
{ "question": "Which segments drove the churn increase last quarter?",
  "tier": "live", "purpose": "churn-analysis-emea", "session_id": "..." }

200 OK
{ "answer": { "headline": "...", "narrative": "...",
              "visual": { "type": "bar", "spec": { ... } },
              "table": { "columns": [...], "rows": [...] } },
  "citations": [ { "product_id": "DP-TEL-001", "contract_version": "3.2.0",
                   "columns": ["segment","churn_flag"], "as_of": "2026-09-03T06:12Z" } ],
  "kpi_definitions": ["KPI-CHURN-001"],
  "trace": { "tool_calls": [...], "rows_scanned": 184203,
             "latency_ms": 4120, "tokens": {"in": 3820, "out": 640},
             "cost_usd": 0.031 },
  "grounded": true, "confidence": 0.91,
  "scope": { "agent_identity": "svc-agent-tel-001",
             "on_behalf_of": "u-48231", "effective_scope": "intersection" } }

422  { "type": "out_of_scope", "detail": "This agent does not cover network fault codes.",
       "suggested_agents": ["AG-TEL-002"], "file_demand_url": "..." }
```

An agent that cannot ground an answer returns a failure with an explanation rather than an ungrounded answer. This is a hard product rule, enforced in the API layer rather than left to the model.

24. Non-Functional Requirements, Security & Compliance

24.1 Performance & Scale

Requirement	Target
Landing page LCP	≤ 1.8 s on 4G; ≤ 1.0 s on broadband
Catalog search response	p95 ≤ 400 ms for a 50k-asset catalog
Detail page render	p95 ≤ 900 ms including quality and adoption panels
Agent answer latency	p95 ≤ 6 s for curated demo questions; ≤ 12 s for free-form
Mesh render	≤ 2 s for 2,000 nodes and 8,000 edges; progressive rendering above that
Catalog scale	50,000 data products, 5,000 agents, 100,000 users, 500 concurrent sessions per tenant
Telemetry ingestion	50M usage events per day per tenant with 15-minute freshness on adoption panels

24.2 Security

- OIDC single sign-on with the enterprise IdP; SCIM provisioning; MFA enforced for owner, steward, architect and administrator roles.
- Least privilege throughout: every connector service principal has a published permission manifest, and CI includes a kill test asserting that no connector can write to a customer platform.
- The marketplace never becomes a shadow authorization store. All data access is provisioned into the platform's native RBAC and revoked there.
- Encryption in transit (TLS 1.3) and at rest; customer-managed keys supported; secrets in a managed vault with rotation.
- Row-level and column-level policy enforcement occurs in the data platform, never in the application layer.
- Agent identities are distinct machine principals; delegated access is always the intersection of agent scope and user entitlement.
- Prompt-injection defences on every retrieval path, with content from untrusted sources treated as data and never as instructions.
- Full audit: every view of a Restricted listing, every request, decision, grant, revocation, agent question and returned column set is logged immutably.

24.3 Availability & Resilience

Aspect	Requirement
Availability	99.9% monthly for the portal and API; 99.5% for the demo console
RPO / RTO	RPO 15 minutes, RTO 4 hours; workflow state is durable and replayable
Degradation	If the data plane is unreachable the catalog remains browsable with last-known metadata, clearly stamped with its as-of time
Agent failure	Agent unavailability degrades to cached curated answers with a visible staleness banner; never to a silent failure

Aspect	Requirement
Backups	Point-in-time recovery for metadata; snapshots retained per the customer retention policy

24.4 Compliance & Tenancy

- **Multi-tenancy** — logical isolation by tenant_id with row-level security at the database, plus optional single-tenant deployment for regulated customers.
- **Residency** — metadata storage region is configurable per tenant; the data plane never leaves the customer's platform.
- **Regulatory posture** — designed to support SOC 2 Type II, ISO 27001, GDPR and CCPA obligations; HIPAA-eligible configuration for healthcare tenants; the EU AI Act transparency obligations are met by the agent listing itself, which documents purpose, data sources, limitations and human oversight for every agent.
- **Retention** — audit events retained for seven years by default; usage telemetry aggregated after 13 months; agent interaction logs retained per tenant policy with configurable redaction.
- **Right to explanation** — for any agent answer, the full chain of question, retrieved context, tool calls, returned columns and answer is reconstructable for the retention period.
- **Accessibility** — WCAG 2.2 AA across all surfaces, with an accessible equivalent for every visualization including both meshes.

25. Seed Catalog — 15 Domain Data Products

This catalog ships with the product. It is sufficient to run a credible client demonstration in any of nine industries on day one, and it doubles as the worked-example library referenced by the academy. Every entry below is fully specified in the seed manifests: contract, columns, quality rules, endpoints, adoption profile and value case.

25.1 Product Register — Purpose & Certified KPIs

ID	Product	Industry / Domain	Grain & purpose	Certified KPIs
DP-TEL-001	Subscriber Churn & Retention 360	Telecom / Customer	One row per subscriber per day, 36 months. Unified view of subscriber state, tenure, plan, usage, service events and churn outcome for retention decisioning.	Churn Rate · Retention Rate · Save Rate · Average Tenure · Churn Propensity Decile
DP-TEL-002	Network Experience & Fault Signal	Telecom / Network & Asset	One row per cell site per hour, 13 months. Radio performance, fault events and customer-impacting degradation joined to service geography.	Network Availability · Dropped Call Rate · Throughput p50/p95 · Fault MTTR · Impacted Subscriber Hours
DP-TCH-001	Product Usage & Feature Adoption	Technology / Product	One row per account per feature per day, 24 months. Instrumented product telemetry mapped to accounts, entitlements and lifecycle stage.	DAU/MAU · Feature Adoption Rate · Activation Rate · Time to Value · Net Revenue Retention
DP-BNK-001	Customer Financial 360	Banking / Customer	One row per customer per month, 60 months. Holdings, balances, channel behaviour, servicing events and relationship value across all lines of business.	Primary Bank Share · Products per Customer · Customer Lifetime Value · Attrition Risk Score · Deposit Balance Growth
DP-BNK-002	Transaction Surveillance & Financial Crime Signal	Banking / Risk & Compliance	One row per alert per disposition event, 84 months. Alerts, typologies, investigation states and outcomes for AML and fraud oversight.	Alert Volume · False Positive Rate · SAR Conversion Rate · Investigation Cycle Time · Backlog Age
DP-INS-001	Claims Lifecycle & Loss Performance	Insurance / Risk	One row per claim per status transition, 84 months. Full claim journey with reserves, payments, adjuster actions and recovery.	Loss Ratio · Claims Cycle Time · Average Severity · Leakage Rate · Recovery Rate
DP-INS-002	Policy & Underwriting Portfolio	Insurance / Product	One row per policy per month, 84 months. In-force portfolio with exposure, pricing, risk attributes and renewal state.	Written Premium · Policy Retention · Quote-to-Bind Rate · Rate Adequacy · Exposure Concentration
DP-HLT-001	Patient Care Journey & Readmission	Healthcare / Clinical	One row per encounter with linked longitudinal journey, 60 months. Admissions, diagnoses, procedures, discharge disposition and follow-up.	30-Day Readmission Rate · Average Length of Stay · ED Utilization Rate · Care Gap Closure · Discharge to Follow-up Interval
DP-HLT-002	Pharmacy & Clinical Supply Utilization	Healthcare / Supply Chain	One row per item per location per day, 36 months. Dispensing, inventory, formulary status and cost for clinical supply.	Formulary Adherence · Stockout Rate · Cost per Patient Day · Waste Rate · Substitution Rate
DP-RTL-001	Omnichannel Sales & Basket Analytics	Retail / Customer	One row per transaction line, 36 months. Sales across store, web and app with basket composition, promotion and margin.	Comparable Sales Growth · Average Basket Value · Conversion Rate · Gross Margin Rate · Promotion Lift
DP-RTL-002	Inventory Position &	Retail / Supply	One row per SKU per location per day, 24 months.	In-Stock Rate · Weeks of Supply · Sell-

ID	Product	Industry / Domain	Grain & purpose	Certified KPIs
	Replenishment Signal	Chain	On-hand, in-transit, demand forecast and replenishment recommendation.	Through Rate · Forecast Accuracy (MAPE) · Lost Sales Estimate
DP-TRN-001	Fleet Movement & Delivery Performance	Transportation / Operations	One row per shipment leg, 36 months. Telematics, route plan versus actual, dwell, exception and cost.	On-Time Delivery Rate · Cost per Mile · Dwell Time · Asset Utilization · Exception Rate
DP-UTL-001	Grid Asset Health & Outage	Utilities / Network & Asset	One row per asset per day plus outage event rows, 84 months. Condition monitoring, maintenance history and interruption events.	SAIDI · SAIFI · Asset Health Index · Average Restoration Time · Preventive Maintenance Compliance
DP-ENG-001	Smart Meter Consumption & Load Profile	Utilities & Energy / Customer	One row per meter per interval, 24 months. Interval consumption, load shape, tariff and demand-response participation.	Peak Demand · Load Factor · Consumption per Customer · DR Event Response Rate · Estimated Read Rate
DP-MFG-001	Manufacturing Yield & Equipment Effectiveness	Manufacturing / Operations	One row per production run per line per shift, 36 months. Availability, performance, quality, downtime reasons and scrap.	OEE · First Pass Yield · Unplanned Downtime Hours · Scrap Rate · Changeover Time

25.2 Product Register — Operational Profile

Freshness and quality figures below are the seed values shipped with the demo tier; in a live deployment they are computed from telemetry and change continuously.

ID	Primary sources	Cadence & SLA	Quality	Sensitivity	Surfaces	Agents
DP-TEL-001	Billing · CRM · Network usage · Care tickets	Daily 06:00, p95 45 min	94 Exemplary	Confidential (PII)	SQL · REST · MCP	AG-TEL-001
DP-TEL-002	RAN OSS · Fault mgmt · Geo reference	Hourly, p95 12 min	88 Healthy	Internal	SQL · MCP · Stream	AG-TEL-001, AG-TEL-002
DP-TCH-001	Product telemetry · CRM · Entitlements · Billing	Hourly, p95 20 min	91 Exemplary	Internal	SQL · REST · MCP	AG-TCH-001
DP-BNK-001	Core banking · Cards · Wealth · Digital channel	Daily 05:00, p95 60 min	96 Exemplary	Restricted (PII)	SQL · MCP · Share	AG-BNK-001
DP-BNK-002	Payments · Sanctions screening · Case mgmt	Every 15 min, p95 6 min	93 Exemplary	Restricted	SQL · MCP	AG-BNK-002
DP-INS-001	Claims admin · Payments · Adjuster notes · Salvage	Daily 04:00, p95 40 min	90 Exemplary	Confidential (PII)	SQL · REST · MCP	AG-INS-001
DP-INS-002	Policy admin · Rating engine · Reinsurance	Daily 04:30, p95 35 min	89 Healthy	Confidential	SQL · MCP	AG-INS-001, AG-INS-002
DP-HLT-001	EHR · ADT feed · Claims · Scheduling	Every 4 hours, p95 25 min	92 Exemplary	Restricted (PHI)	SQL · MCP	AG-HLT-001
DP-HLT-002	Pharmacy system · ERP inventory · Formulary	Daily 03:00, p95 30 min	87 Healthy	Confidential	SQL · REST · MCP	AG-HLT-002

ID	Primary sources	Cadence & SLA	Quality	Sensitivity	Surfaces	Agents
DP-RTL-001	POS · E-commerce · Loyalty · Promotions	Hourly, p95 15 min	95 Exemplary	Internal	SQL · REST · MCP	AG-RTL-001
DP-RTL-002	WMS · ERP · Demand forecast · Store ops	Every 2 hours, p95 18 min	89 Healthy	Internal	SQL · MCP · Stream	AG-RTL-001, AG-RTL-002
DP-TRN-001	TMS · Telematics · ELD · Fuel cards	Every 30 min, p95 8 min	86 Healthy	Internal	SQL · MCP · Stream	AG-TRN-001
DP-UTL-001	SCADA · OMS · Asset registry · Work mgmt	Every 15 min, p95 7 min	91 Exemplary	Internal	SQL · MCP · Stream	AG-UTL-001
DP-ENG-001	AMI head-end · Billing · Tariff reference	Daily 02:00, p95 50 min	88 Healthy	Confidential (PII)	SQL · MCP	AG-UTL-001
DP-MFG-001	MES · Historian · Quality system · CMMS	Per shift close, p95 20 min	90 Exemplary	Internal	SQL · REST · MCP	AG-MFG-001

25.3 Seed Value Cases

Each seed product ships with a complete value case in the Section 18 structure. Headline benefit models, with all assumptions stated in the manifest:

ID	Business outcome	Benefit model headline
DP-TEL-001	Retention teams target save offers at subscribers genuinely at risk instead of by tenure band	Save-offer precision improvement × contribution margin per retained subscriber, less offer cost
DP-TEL-002	Field engineering prioritizes sites by customer impact rather than by alarm count	Impacted subscriber-hours avoided × churn elasticity per impacted hour
DP-TCH-001	Customer success intervenes on adoption stall before renewal risk materializes	At-risk ARR identified early × historical save rate on early intervention
DP-BNK-001	Relationship managers see whole-customer value across lines of business before pricing decisions	Cross-sell conversion uplift + attrition avoided on high-value households
DP-BNK-002	Investigators work the alerts most likely to convert instead of the queue in arrival order	Investigator hours redirected × loaded rate + regulatory backlog risk reduction
DP-INS-001	Claims leadership sees leakage drivers by adjuster, coverage and cycle stage within days, not quarters	Leakage basis points recovered × claims paid
DP-INS-002	Underwriting sees rate adequacy and concentration before binding rather than at quarter end	Loss-ratio improvement on repriced segments + capital efficiency from concentration control
DP-HLT-001	Care management targets discharge follow-up at patients with genuine readmission risk	Readmissions avoided × average penalty and cost per readmission
DP-HLT-002	Pharmacy reduces stockouts and waste simultaneously by seeing utilization and inventory together	Waste reduction + substitution avoidance + stockout-driven expedite cost avoided
DP-RTL-001	Merchandising decisions move from weekly reporting cycles to same-day	Margin improvement from faster markdown and assortment decisions

ID	Business outcome	Benefit model headline
DP-RTL-002	Replenishment responds to real demand signal rather than to average weekly movement	Lost sales recovered × margin, less carrying cost of the added position
DP-TRN-001	Operations sees the cost and service consequence of routing decisions on the same screen	On-time improvement × penalty avoidance + cost per mile reduction
DP-UTL-001	Maintenance is scheduled by asset condition and customer impact rather than by calendar	SAIDI minutes avoided × regulatory value per customer-minute + deferred replacement capital
DP-ENG-001	Demand response and tariff design are grounded in actual load shape by segment	Peak demand reduction × avoided capacity cost + DR programme yield
DP-MFG-001	Plant leadership sees the true driver of OEE loss by line and shift within one shift	Downtime hours recovered × contribution per production hour + scrap reduction

26. Seed Catalog — 14 AI Agents

Every seed agent ships with a complete coverage map, an entitlement scope, an evaluation set, a value case, and the five curated demo exchanges specified in Section 27. Autonomy levels follow Section 10: L0 answers only, L1 answers with recommendations, L2 drafts actions for human approval, L3 executes scoped actions with audit.

26.1 Agent Register

ID	Agent	Industry	Capability statement	Data products	L
AG-TEL-001	Churn Sentinel	Telecom	Answers questions about subscriber churn, retention performance and save-offer targeting, and explains what is driving churn movement.	DP-TEL-001, DP-TEL-002	L1
AG-TEL-002	Network Quality Advisor	Telecom	Answers questions about network availability, degradation and fault resolution, ranked by customer impact rather than alarm volume.	DP-TEL-002	L1
AG-TCH-001	Product Adoption Analyst	Technology	Answers questions about feature adoption, activation, time to value and accounts whose usage signals renewal risk.	DP-TCH-001	L1
AG-BNK-001	Relationship Value Advisor	Banking	Answers questions about household relationship value, product holdings, attrition risk and cross-sell opportunity.	DP-BNK-001	L1
AG-BNK-002	Financial Crime Triage	Banking	Answers questions about alert volume, false positives, SAR conversion and backlog, and drafts a prioritized investigation queue for approval.	DP-BNK-002	L2
AG-INS-001	Claims Leakage Investigator	Insurance	Answers questions about loss ratio, claims cycle time, severity and leakage, and ranks the drivers behind each movement.	DP-INS-001, DP-INS-002	L1
AG-INS-002	Underwriting Portfolio Copilot	Insurance	Answers questions about rate adequacy, retention, quote-to-bind and exposure concentration across the in-force portfolio.	DP-INS-002	L1
AG-HLT-001	Readmission Risk Navigator	Healthcare	Answers operational questions about readmission rates, length of stay, discharge follow-up and care-transition programme performance.	DP-HLT-001	L1
AG-HLT-002	Pharmacy Utilization Agent	Healthcare	Answers questions about formulary adherence, stockout risk, waste and pharmacy cost, and drafts replenishment adjustments for approval.	DP-HLT-002	L2
AG-RTL-001	Merchandising Performance Agent	Retail	Answers questions about comparable sales, basket, margin and promotion performance, and explains category variance to plan.	DP-RTL-001, DP-RTL-002	L1
AG-RTL-002	Inventory Availability Agent	Retail	Answers questions about in-stock rate, weeks of supply, forecast accuracy and lost sales, and drafts transfer recommendations for approval.	DP-RTL-002	L2
AG-TRN-001	Delivery SLA Sentinel	Transportation	Answers questions about on-time delivery, dwell, utilization and cost per mile, and identifies customers at risk of SLA penalty.	DP-TRN-001	L1

ID	Agent	Industry	Capability statement	Data products	L
AG-UTL-001	Outage Impact Analyst	Utilities & Energy	Answers questions about SAIDI, SAIFI, asset health and outage impact, and ranks maintenance by customer-minute risk.	DP-UTL-001, DP-ENG-001	L1
AG-MFG-001	Yield & Downtime Analyst	Manufacturing	Answers questions about OEE, first pass yield, downtime reasons and scrap, and decomposes losses by line, shift and product.	DP-MFG-001	L1

26.2 Coverage, Scope & Value Profile

ID	KPIs covered	Explicitly out of scope	Seed value headline
AG-TEL-001	Churn Rate · Retention Rate · Save Rate · Tenure · Propensity Decile	Individual subscriber credit decisions; pricing approval; anything requiring billing write-back	~310 analyst hours deflected per month at 91% acceptance
AG-TEL-002	Availability · Dropped Call Rate · Throughput · Fault MTTR · Impacted Subscriber Hours	Radio parameter changes; work-order creation; capital planning approval	~180 hours deflected; median fault triage time down from 40 to 9 minutes
AG-TCH-001	DAU/MAU · Feature Adoption · Activation · Time to Value · NRR	Contract or pricing commitments; individual user behaviour without consent basis	~240 hours deflected; 22% of at-risk ARR surfaced earlier than the prior process
AG-BNK-001	Primary Bank Share · Products per Customer · CLV · Attrition Risk · Balance Growth	Credit decisions; suitability advice; any individual customer recommendation without banker review	~420 hours deflected across 1,100 relationship managers
AG-BNK-002	Alert Volume · False Positive Rate · SAR Conversion · Cycle Time · Backlog Age	Filing a SAR; closing an alert; naming a customer as suspicious	Investigator effort redirected to alerts with 3.1× higher conversion
AG-INS-001	Loss Ratio · Cycle Time · Severity · Leakage Rate · Recovery Rate	Coverage determination; settlement authority; individual adjuster performance ratings	~260 hours deflected; leakage drivers identified within days of month close
AG-INS-002	Written Premium · Retention · Quote-to-Bind · Rate Adequacy · Concentration	Binding authority; rate filing; individual risk acceptance	~200 hours deflected; concentration breaches surfaced pre-bind
AG-HLT-001	30-Day Readmission · ALOS · ED Utilization · Care Gap Closure · Follow-up Interval	Any individual clinical recommendation, diagnosis or treatment guidance; patient-level advice	~190 hours deflected across care management and quality teams
AG-HLT-002	Formulary Adherence · Stockout Rate · Cost per Patient Day · Waste Rate · Substitution	Clinical substitution decisions; ordering without pharmacist approval	~150 hours deflected; drafted replenishments accepted at 78%
AG-RTL-001	Comp Sales · Basket Value · Conversion · Margin Rate · Promotion Lift	Price changes; markdown execution; vendor negotiation positions	~380 hours deflected; category reviews shortened from 3 days to same-day
AG-RTL-002	In-Stock Rate · Weeks of Supply · Sell-Through · Forecast MAPE · Lost Sales	Executing transfers or purchase orders without planner approval	~290 hours deflected; drafted transfers accepted at 71%
AG-TRN-001	On-Time Delivery · Cost per Mile · Dwell · Utilization · Exception Rate	Dispatch changes; carrier contracting; driver performance actions	~210 hours deflected; penalty exposure identified 6 days earlier on average
AG-UTL-001	SAIDI · SAIFI · Asset Health Index ·	Switching orders; crew dispatch; regulatory filings	~230 hours deflected; storm impact

ID	KPIs covered	Explicitly out of scope	Seed value headline
	Restoration Time · PM Compliance		assessment reduced from 2 days to 3 hours
AG-MFG-001	OEE · First Pass Yield · Unplanned Downtime · Scrap Rate · Changeover Time	Equipment parameter changes; maintenance scheduling without planner approval	~270 hours deflected; shift-level loss attribution available at shift close

26.3 Publish Gate for Agents

An agent cannot be published until every item below passes. The gate runs in CI and its result is displayed on the agent detail page.

- Capability statement, business-value block and explicit out-of-scope list all present and non-trivial.
- Coverage map complete: every KPI row cites an existing certified definition ID and names a source product and column set.
- At least five curated demo exchanges, each validated against a golden answer within tolerance in the last seven days.
- Entitlement scope reviewed and approved; classification coverage verified — no reachable column carries a tag the scope does not permit.
- Compositional exposure check passed where the agent invokes other agents.
- Evaluation suite present with a documented pass threshold; current run above threshold.
- Grounding check: no evaluation case produces an uncited numeric claim.
- Owner, on-call rotation and escalation path recorded; value case authored with assumptions stated.

27. Demo Question Bank — 70 Scripted Exchanges

Five exchanges per agent, authored as data in the agent manifest and validated nightly against golden answers. Headlines below are illustrative of the shape and specificity required; the live values come from the demo-tier dataset. Every answer renders with the KPI definition used, an as-of timestamp, and citations to the source products.

AG-TEL-001 — Churn Sentinel

#	Scripted question	Illustrative answer headline	Analysis · visual
1	What is our churn rate this quarter and how does it compare to last?	Churn is 1.72% quarter to date, up 0.28 pp on Q2. The increase is concentrated in the 13–18 month tenure band, which accounts for 63% of the movement.	Period comparison · line + delta callout
2	Which customer segments drove the churn increase last quarter?	Postpaid family plans in the Midwest contributed 47% of the increase, rising 3.1 pp after the June plan change; every other segment moved less than 0.4 pp.	Driver ranking · waterfall
3	Are network issues contributing to churn in the Midwest?	Subscribers with more than four service-impacted hours in the prior 30 days churn at 2.9× the regional baseline, and 61% of them are served by just 12 degraded sites.	Correlation · scatter + site table
4	Which subscribers should we prioritise for a save offer this week?	8,412 subscribers sit in the top propensity decile with contract expiry inside 30 days and lifetime value above $1,100 — 71% higher expected save value than the current tenure-based list.	Cohort targeting · ranked table
5	Did the May retention campaign actually work?	The treated cohort churned at 1.31% against 1.86% for a matched control over 90 days — a 0.55 pp save at $34 cost per subscriber retained.	Cohort comparison · dual line

AG-TEL-002 — Network Quality Advisor

#	Scripted question	Illustrative answer headline	Analysis · visual
1	Which sites had the worst customer impact yesterday?	Ten sites account for 68% of impacted subscriber-hours, led by three Denver-cluster sites at 14,200 hours combined — all showing sector congestion rather than hardware fault.	Impact ranking · bar + map
2	What is driving the throughput drop in the Denver cluster?	p50 throughput fell 34% between 18:00 and 21:00 on three sectors; backhaul utilization exceeded 92% in the same window while radio availability stayed at 99.8%.	Root cause · time-series overlay
3	How does availability compare across regions this month?	Availability ranges from 99.94% in the Northeast to 99.61% in the Mountain region; the gap is entirely explained by two multi-day fibre events.	Comparison · small multiples
4	Which faults are recurring on the same assets?	23 sites logged three or more faults of the same class within 60 days, concentrated in two equipment generations — a repeat-fault pattern that preventive replacement would address.	Pattern detection · table
5	What is our MTTR trend and where is it worst?	MTTR improved from 4.8 to 3.9 hours over six months, but rural Mountain-region faults remain at 9.2 hours, driven by travel time rather than diagnosis.	Trend + segmentation · line + box

AG-TCH-001 — Product Adoption Analyst

#	Scripted question	Illustrative answer headline	Analysis · visual
1	Which features are driving activation this quarter?	Three features appear in 82% of accounts that activated within 14 days; accounts that skip the second of them activate 3.2× slower.	Driver ranking · bar + funnel
2	Which accounts show adoption decline ahead of renewal?	47 accounts representing $4.1M ARR show a 30%+ decline in weekly active users over 60 days with renewal inside 120 days.	Risk cohort · ranked table
3	How long does a new account take to reach first value?	Median time to value is 11 days, but the enterprise segment sits at 26 days — driven by SSO configuration, which blocks 41% of enterprise onboardings.	Distribution · histogram + segment split
4	Did the March onboarding change improve activation?	Activation within 14 days rose from 44% to 58% for cohorts after the change, holding across segments; the effect is stable over four monthly cohorts.	Cohort comparison · cohort curve
5	Which features have high usage but low entitlement coverage?	Four features are used by accounts on plans that do not include them, worth an estimated $2.3M in expansion if converted to paid entitlement.	Gap analysis · matrix

AG-BNK-001 — Relationship Value Advisor

#	Scripted question	Illustrative answer headline	Analysis · visual
1	Who are my top 20 households by relationship value in the Southeast?	The top 20 households hold $187M in combined balances and average 4.2 products; six of them have declining primary-bank share.	Ranking · table with trend sparklines
2	Which high-value customers show attrition risk this month?	312 households above $250K in balances moved into the elevated attrition band, most commonly after a direct-deposit redirection event.	Risk cohort · ranked table
3	What is our primary bank share trend by segment?	Primary-bank share is flat overall at 61% but has fallen 4 pp in the 25–34 segment over 12 months while rising 2 pp in 55+.	Trend by segment · line
4	Which single-product customers show strong balance growth?	4,180 customers hold only a deposit relationship yet grew balances more than 25% year over year — the highest-yield cross-sell pool in the book.	Opportunity cohort · table
5	How did deposit balances shift after the April rate change?	Balances fell 2.1% overall in the following 60 days, with 78% of the outflow from rate-sensitive households already flagged in the attrition band.	Event impact · before/after line

AG-BNK-002 — Financial Crime Triage

#	Scripted question	Illustrative answer headline	Analysis · visual
1	What is the alert backlog and its age profile?	4,120 alerts are open; 640 exceed the 30-day SLA, concentrated in two typologies that grew after the June rule change.	Backlog profile · stacked age bars
2	Which typologies have the highest false positive rate?	Structuring-pattern alerts convert at 1.2% against a 4.8% portfolio average, producing 38% of investigator hours for 9% of outcomes.	Efficiency ranking · bar
3	Which alerts should be worked first today?	A ranked queue of 180 alerts with 3.1× the baseline conversion likelihood, drafted for supervisor approval before assignment.	Prioritization · ranked queue (drafted, requires approval)
4	Has SAR conversion changed since the June rule tuning?	Conversion rose from 3.9% to 5.6% while alert volume fell 22%, with no measurable reduction in downstream confirmed cases.	Before/after · dual axis

#	Scripted question	Illustrative answer headline	Analysis · visual
5	Which investigators are carrying aged alerts above SLA?	Nine investigators hold 71% of the over-SLA backlog; the concentration follows two specialised typologies rather than individual throughput.	Workload distribution · table

AG-INS-001 — Claims Leakage Investigator

#	Scripted question	Illustrative answer headline	Analysis · visual
1	Where is leakage highest by coverage line?	Auto physical damage shows an estimated 4.1% leakage rate against a 2.3% book average, worth $18.4M annualized.	Ranking · bar with value
2	Which settlement patterns diverge most from peers on like claims?	Three handling patterns settle comparable-complexity claims 12–18% above the matched peer band; the variance concentrates in one office and one claim type.	Variance analysis · box plot
3	What is driving cycle time increase in auto physical damage?	Median cycle time rose 3.4 days, 79% attributable to the appraisal step following a vendor change in April.	Decomposition · waterfall by stage
4	How much subrogation recovery are we leaving on the table?	$6.2M in claims met referral criteria but were never referred, 62% of them below the manual review threshold.	Gap quantification · funnel
5	Did the new triage rule reduce severity on low-complexity claims?	Average severity on triaged claims fell 7.8% with cycle time down 2.1 days and no increase in reopen rate over 120 days.	Programme evaluation · before/after

AG-INS-002 — Underwriting Portfolio Copilot

#	Scripted question	Illustrative answer headline	Analysis · visual
1	Where is rate adequacy weakest in the commercial property book?	Three segments price below indicated by 8–14%, together representing 22% of written premium and 31% of incurred loss.	Adequacy ranking · scatter
2	What is our exposure concentration by county for wind peril?	Two coastal counties carry 19% of total insured value against a 12% guideline, driven by growth in one distribution channel.	Concentration · map + table
3	How has quote-to-bind moved since the July rate change?	Quote-to-bind fell from 24% to 19% overall, but the decline is confined to the two segments that took the largest increases.	Trend by segment · line
4	Which segments are we retaining below plan?	Four segments sit 5–11 pp below retention plan; three of them also show the largest rate increases, indicating price-driven attrition.	Variance to plan · bar
5	What would a 5% increase on the worst-performing segment do to retention?	Applying the observed historical elasticity, retention would fall an estimated 3.2 pp — a directional projection, not a pricing recommendation.	Scenario framing · elasticity curve (directional)

AG-HLT-001 — Readmission Risk Navigator

#	Scripted question	Illustrative answer headline	Analysis · visual
1	What is our 30-day readmission rate and how does it compare to last year?	The rate is 14.2%, down 0.9 pp year over year, with the improvement concentrated in two service lines.	Trend comparison · line
2	Which conditions and units drive the most	Four condition groups account for 58% of readmissions; two medical units	Contribution ranking ·

#	Scripted question	Illustrative answer headline	Analysis · visual
	readmissions?	contribute disproportionately relative to their volume.	Pareto
3	Which discharge patterns are associated with higher readmission?	Discharges without a scheduled follow-up inside seven days readmit at 19.4% against 11.1% when follow-up is scheduled.	Association analysis · comparison bars
4	Are follow-up appointments being scheduled within the target window?	68% of discharges have follow-up scheduled within seven days against a 90% target; the shortfall concentrates in weekend discharges.	Compliance · gauge + segment split
5	Did the transitional care programme reduce readmissions in the pilot units?	Pilot units fell from 15.8% to 12.4% over six months against 15.1% in matched comparison units.	Programme evaluation · dual line

AG-HLT-002 — Pharmacy Utilization Agent

#	Scripted question	Illustrative answer headline	Analysis · visual
1	Which items are at risk of stockout this week?	34 items fall below safety stock against forecast demand, 11 of them single-source with lead times beyond the coverage window.	Risk list · ranked table
2	Where is formulary adherence lowest?	Adherence is 82% overall but 61% in two units, driven by three therapeutic classes with available formulary equivalents.	Compliance ranking · bar
3	What is driving the increase in pharmacy cost per patient day?	Cost per patient day rose 6.4%; 71% of the increase comes from two high-cost classes where non-formulary substitution rose.	Decomposition · waterfall
4	How much waste came from short-dated stock last quarter?	$412K in doses expired unused, 58% concentrated in four items with over-ordering against declining utilization.	Waste analysis · Pareto
5	Draft a replenishment adjustment for the top 10 at-risk items.	A drafted adjustment covering 10 items with quantities, rationale and cost impact, routed to the pharmacy manager for approval before any order is placed.	Action draft · approval card (L2)

AG-RTL-001 — Merchandising Performance Agent

#	Scripted question	Illustrative answer headline	Analysis · visual
1	How are comparable sales trending by category this week?	Comp sales are up 3.1% overall; two categories are negative, with home decor down 4.2% against a positive prior four weeks.	Trend by category · heatmap
2	Which categories are missing plan and why?	Three categories miss plan by more than 5%; for two, availability explains most of the gap, and for the third, traffic conversion does.	Variance decomposition · waterfall
3	Did the back-to-school promotion deliver incremental margin?	The promotion lifted units 18% but margin dollars only 4% after markdown and halo effects; two of five promoted SKUs were margin-dilutive.	Promotion evaluation · lift analysis
4	Which SKUs have high sell-through but poor in-stock?	112 SKUs exceed 85% sell-through while sitting below 90% in-stock, representing an estimated $2.8M in recoverable sales.	Opportunity matrix · quadrant
5	Where is basket size declining and what is leaving the basket?	Basket value fell 2.4% in the urban store cluster, driven almost entirely by attachment items dropping out of grocery-anchored trips.	Basket composition · affinity shift

AG-RTL-002 — Inventory Availability Agent

#	Scripted question	Illustrative answer headline	Analysis · visual
1	Which stores have the worst in-stock rate this week?	18 stores sit below 92% in-stock; 14 share the same distribution centre and the same three replenishment exceptions.	Ranking · table + map
2	What is our forecast accuracy by category?	Portfolio MAPE is 18.4%; two seasonal categories exceed 30%, which explains most of the current over-position.	Accuracy ranking · bar
3	Where do we have excess weeks of supply?	$14.2M of inventory sits above 12 weeks of supply, 62% concentrated in three categories following a forecast overshoot.	Excess analysis · Pareto
4	What did stockouts cost us last month?	Estimated lost sales of $3.6M, with 44% recoverable through transfers from stores holding excess of the same SKU.	Lost sales · quantified estimate
5	Draft transfer recommendations to fix the top 20 availability gaps.	Twenty drafted transfers with source, destination, quantity, freight cost and expected recovered sales, routed to the planner for approval.	Action draft · approval card (L2)

AG-TRN-001 — Delivery SLA Sentinel

#	Scripted question	Illustrative answer headline	Analysis · visual
1	What is on-time delivery this week and where is it worst?	On-time delivery is 93.1% against a 96% target; the Northeast lanes sit at 87.4% and account for 71% of the total miss.	Performance ranking · bar + map
2	What is driving late deliveries on the Northeast lanes?	Origin dwell above 90 minutes precedes 64% of late arrivals, concentrated at two facilities during the evening dock window.	Root cause · contribution analysis
3	Where is dwell time eroding asset utilization?	Excess dwell consumed 4,180 tractor-hours last month — the equivalent of 11 tractors — concentrated at six facilities.	Utilization impact · quantified
4	How does cost per mile compare across carriers on the same lanes?	On 22 matched lanes, cost per mile ranges from $2.14 to $2.87 with no corresponding service difference.	Like-for-like comparison · scatter
5	Which customers are at risk of SLA penalty this month?	Seven customer contracts are tracking below their service threshold with 11 days remaining, carrying $340K of exposure.	Risk forecast · ranked table

AG-UTL-001 — Outage Impact Analyst

#	Scripted question	Illustrative answer headline	Analysis · visual
1	What is our SAIDI year to date against the regulatory target?	SAIDI stands at 98 minutes against a 110-minute target, but excluding major event days the underlying trend is flat rather than improving.	Regulatory tracking · line vs target
2	Which feeders contribute most to interruption minutes?	Twelve feeders produce 41% of customer-interruption minutes; nine share the same vegetation-management cycle.	Contribution ranking · Pareto
3	Which poor-health assets serve the most customers?	38 assets in the lowest health band each serve more than 2,000 customers, together representing 214,000 customers of exposure.	Risk prioritization · matrix
4	What was the customer and load impact of last week's storm?	84,000 customers interrupted for a combined 1.9M customer-minutes, with 62% restored inside four hours and load recovery complete within 18.	Event analysis · timeline + map

#	Scripted question	Illustrative answer headline	Analysis · visual
5	Which maintenance deferrals carry the highest customer-minute risk?	Nine deferred work orders carry an estimated 180,000 customer-minutes of annual risk, five of them on assets already in the lowest health band.	Risk ranking · table

AG-MFG-001 — Yield & Downtime Analyst

#	Scripted question	Illustrative answer headline	Analysis · visual
1	What is OEE by line this week and where did we lose the most?	Plant OEE is 71.4%; Line 3 sits at 58.2% and accounts for 44% of the total loss, almost entirely on the availability component.	Performance ranking · OEE waterfall
2	What are the top downtime reasons on Line 3?	Three reason codes explain 68% of unplanned downtime, led by a changeover fault recurring on one product transition.	Pareto · reason code ranking
3	Which shifts show the biggest first-pass-yield gap?	Night shift first-pass yield trails day shift by 4.8 pp on the same products and equipment, with the gap concentrated in the first two hours.	Shift comparison · box plot
4	Did changeover standardization reduce changeover time?	Median changeover fell from 47 to 31 minutes on standardized transitions, with variance down more than the median — the more valuable result.	Programme evaluation · distribution shift
5	Where is scrap concentrated by product and root cause?	Two products contribute 57% of scrap value; for one, a single upstream dimensional cause accounts for most of it.	Concentration · treemap + Pareto

28. Certified KPI Registry & Semantic Backbone

Everything else in this specification depends on one idea: a business measure has exactly one authoritative definition, and every product, agent, chart and answer cites it. Without that, the catalog is a list, the agent mesh cannot detect divergence, and two agents can give a board two different churn rates with equal confidence. The KPI registry is therefore not a feature of the semantic layer — it is the spine the marketplace is built on.

28.1 Definition Record

```
kpi:
  id: KPI-CHURN-001
  name: "Subscriber Churn Rate"
  status: certified                 # draft | certified | deprecated | superseded
  owner: { steward: "A. Vance", forum_approved: 2026-04-18 }
  business_definition: >
    The share of subscribers active at the start of a period who have
    disconnected all services by the end of that period.
  calculation:
    numerator:   "count(distinct subscriber_id where disconnect_date within period)"
    denominator: "count(distinct subscriber_id active at period_start)"
    grain_supported: [day, week, month, quarter]
    slices_supported: [segment, region, plan_type, tenure_band, channel]
  inclusions:  ["voluntary and involuntary disconnects"]
  exclusions:  ["plan migrations within the same account", "device-only accounts"]
  unit: percentage   direction: lower_is_better   target: 1.45
  source_of_record: DP-TEL-001
  synonyms: ["attrition rate", "churn %", "subscriber loss rate"]
  related: { conflicts_with: [], superseded_by: null }
  review: { cadence_months: 6, last_reviewed: 2026-08-02 }
```

28.2 Registry Rules

- **One active definition per name per tenant.** Publishing a second is a blocking error with a side-by-side diff and a required resolution: adopt, rename, or supersede. There is no "approve anyway" path.
- **Nothing computes an uncited measure.** A product cannot declare a certified KPI without a definition ID; an agent cannot answer a KPI it does not cite; a demo exchange cannot pass validation without a must_cite entry. The registry is enforced at three gates, not by convention.
- **Definitions are versioned and their consumers are known.** Changing a calculation creates a new version, notifies every product, agent, report and value case that cites it, and annotates every trend chart at the change date so a step in the line is never mistaken for a business event.
- **Synonyms are first-class.** Universal search, the intake duplicate matcher and agent natural-language understanding all read the same synonym list. A synonym added once is understood everywhere.
- **Deprecation has a parallel-run period.** A superseded definition remains queryable for the notice window with a visible banner, so downstream reconciliation is possible.

28.3 Divergence Detection

A registry that is only enforced at publish time drifts silently in production. A scheduled reconciliation job asks the same question of every agent and every semantic model that claims a KPI, over the same period and slice, and compares.

Check	Frequency	Failure handling
Cross-agent agreement on a shared KPI	Daily, per KPI answered by more than one agent	Variance beyond tolerance opens an incident against both agents and banners both listings until resolved
Product-to-product agreement on a shared KPI	Daily, per KPI served by more than one product	Incident against both products; the mesh shared-KPI edge is flagged
Semantic layer conformance	On every semantic model deploy	A measure whose expression no longer matches its definition blocks the deploy
BI reconciliation	Weekly, against connected BI tools	A dashboard measure diverging from the certified definition is raised as a curation finding, not silently overwritten
Orphan measure detection	Weekly	A measure computed in a product or report with no registry entry is queued for steward triage

29. AgentOps — Evaluation, Safety & Release Governance

An agent is a released software artifact whose behaviour changes when a model, prompt, tool binding or upstream dataset changes — four moving parts, three of which are outside the agent team's control. The platform therefore treats agent releases with the discipline normally reserved for production services.

29.1 The Agent Release Bundle

An agent version is immutable and pins every input that can alter behaviour. Rolling back is restoring a bundle, never re-tuning a prompt in production.

Pinned element	Detail
Model	Provider, model identifier, version and inference parameters (temperature, top-p, max tokens, seed where supported)
Prompts	System prompt, tool descriptions and few-shot content, content-hashed and stored in the prompt registry
Tool bindings	Exact MCP tool set with scope strings, row limits and cost classes
Data contracts	Contract version of every upstream product at publish time
KPI definitions	Definition version for every KPI in the coverage map
Guardrail config	Refusal policy, output filters, injection defences and their versions
Evaluation result	The full evaluation run that authorised the release, with pass rate and per-suite detail

29.2 Evaluation Suites

Suite	What it proves	Gate
Golden accuracy	The agent returns the correct value for known questions with known answers, within tolerance, across every KPI in the coverage map	Blocking. Minimum 20 cases per KPI class; pass threshold declared per agent and never lowered to pass a release
Groundedness	Every numeric claim carries a valid citation resolving to a real product, column set and as-of timestamp	Blocking at 100%. An uncited number is a defect, not a warning
Boundary and refusal	Out-of-scope questions are refused with a named boundary and a handoff, not answered speculatively	Blocking. Includes the explicit out-of-scope list from the agent manifest
Adversarial and injection	Prompt injection through retrieved content, tool output and user input fails to alter scope, exfiltrate columns or bypass purpose binding	Blocking. Suite maintained centrally and versioned; every agent runs the current version
Entitlement enforcement	Under a low-privilege delegated identity the agent returns strictly less, never more, and never leaks the existence of masked columns	Blocking. Run against at least three synthetic personas of differing entitlement
Compositional exposure	Where the agent invokes other agents or composes tools, the union of reachable data does not exceed the approved scope	Blocking where composition exists
Consistency	The same question asked five times returns materially the same answer	Advisory with a published variance figure; high variance blocks certification but not

Suite	What it proves	Gate
		publication
Cost and latency	p95 latency and cost per answer stay within the declared budget on the evaluation corpus	Blocking against the budget in the agent manifest

29.3 Release Path

13. **Author.** Changes are made to the manifest and prompt registry in a branch. Nothing is edited in a running agent.
14. **Evaluate.** The full suite runs against the demo tier. Results, including per-case diffs against the previous version, are attached to the pull request.
15. **Review.** Owner plus steward review; a security review is required when scope, tool bindings or model provider change.
16. **Canary.** The new version serves a defined share of traffic — 10% by default — with live acceptance rate, groundedness, latency and cost compared to the incumbent. Minimum soak of 200 answered questions or 48 hours, whichever comes first.
17. **Promote or roll back.** Promotion requires no regression on any blocking suite and no degradation beyond tolerance on canary metrics. Rollback is a single action restoring the previous bundle and is exercised in a quarterly drill.
18. **Record.** The release, its evaluation evidence and its approver are written to the immutable audit log and rendered on the agent's Safety tab.

29.4 Runtime Safety Controls

Control	Behaviour
Untrusted content isolation	Retrieved documents, tool results and user text are delivered to the model as data with explicit provenance framing and are never treated as instructions. Tool-call decisions derived solely from retrieved content are blocked and logged.
Purpose enforcement	Every tool call carries the purpose bound at grant time; a call without a valid purpose fails closed at the MCP gateway rather than at the model.
Output filtering	Sensitive-pattern detection on generated output as a defence in depth. A hit is a security event, and it is also treated as evidence that a masking policy upstream is missing.
Blast-radius limits	Row limits, cost classes and per-session call ceilings per tool. L2 agents draft only; L3 agents execute only reversible, scoped actions with an approval record and an undo path.
Kill switch	An owner, steward or security officer can disable an agent instantly. Disabled agents remain visible with a status banner rather than vanishing, so consumers understand what happened.
Red teaming	Quarterly structured exercise per L2 and L3 agent, and on any material scope change. Findings become permanent cases in the adversarial suite.

29.5 Documentation Obligations

The agent listing is the transparency artifact. Purpose, data sources, limitations, out-of-scope statement, evaluation results, human oversight arrangement, autonomy level and owner are all published on it and versioned with the agent. This is deliberately the same content that a regulator, an internal model-risk committee or a curious business user would ask for, so there is one artifact rather than three that drift apart.

30. Cost Governance & FinOps

Agents make cost per question a live operational metric rather than an annual platform line item. A marketplace that grows adoption without unit-economic visibility will succeed its way into a budget crisis.

30.1 Cost Model

Cost element	Attribution	Rendered on
Warehouse compute	Per query, tagged with product, consumer, purpose and surface (SQL, REST, MCP, agent)	Product consumption tab, department showback
Storage	Per product including history depth and demo-tier copies	Product cost of ownership in the value case
Model inference	Per answer: input and output tokens, model, tool-call count	Agent observability and value tabs
Retrieval and vector	Embedding refresh and search invocation per corpus product	Product cost of ownership
Platform	Marketplace infrastructure allocated per active asset	Portfolio view only, never per consumer
Stewardship	Recorded effort hours per asset from the certification and backlog workflow	Value case cost of ownership

30.2 Unit Economics

```
cost_per_answer          = inference_cost + retrieval_cost + query_cost   (per agent answer)
cost_per_accepted_answer = cost_per_answer / answer_acceptance_rate
cost_per_active_consumer = total_asset_cost / distinct_consumers_90d      (per data product)
value_ratio              = realized_value / total_cost_of_ownership

Standing targets, stored as rubric data and reviewed quarterly:
  cost_per_accepted_answer  <= 0.35 × loaded cost of the manual alternative
  value_ratio               >= 3.0 for certified assets after four quarters
  demo-tier cost            <= 5% of total agent inference spend
```

30.3 Controls

- **Budgets and alerts per asset and per department**, with a soft threshold that notifies the owner and a hard threshold that throttles non-interactive workloads before it throttles interactive ones.
- **Cost classes on every MCP tool** so an agent cannot accidentally invoke a full-scan tool in a loop; the class is declared in the manifest and enforced at the gateway.
- **Anomaly detection on cost per query** with volume held constant, which catches inefficient rewrites and runaway retries earlier than any monthly bill review.
- **Retirement economics.** The portfolio view ranks assets by cost with no offsetting adoption; a product with zero queries in 90 days and non-trivial storage is proposed for deprecation automatically, with the saving quantified.
- **Chargeback is optional, showback is not.** Every department can see what it consumed whether or not the organisation charges for it, because visibility alone changes behaviour.

31. Standards & Interoperability

Every interface in this platform is an existing open standard where one exists. Proprietary interfaces are a liability at enterprise scale: they raise integration cost, complicate security review, and make the platform harder to displace for the wrong reason. Conformance is asserted by tests, not claimed in marketing.

Standard	Used for	Conformance test
Open data product / data contract specification	Product manifest and contract structure, with platform extensions namespaced separately	Manifests validate against the published schema in CI; extensions live under a reserved key
OpenLineage	Lineage ingestion from pipelines and emission of marketplace-derived lineage	Round-trip test: emitted events re-ingest without loss
Model Context Protocol	The AI consumption surface for every data product and the tool binding surface for agents	Generated servers pass an MCP conformance suite; a third-party client consumes a product with no bespoke code
Agent interoperability descriptors	Publishing agent capability, scope and invocation metadata for cross-platform discovery	Agent card generated from the manifest and validated; kept optional until the standard stabilises
OpenAPI 3.1	REST surface and generated SDKs	Spec generated from source; drift between code and spec fails the build
OpenTelemetry	Traces, metrics and logs across portal, API, MCP gateway and platform query	One trace ID spans the full path in an end-to-end test
OIDC and SCIM	Authentication and user or group provisioning	Integration test against a reference IdP
OAuth 2.0 with delegated identity	Agent and application access, purpose-scoped	Scope escalation and missing-purpose cases fail closed in the security suite
DCAT-family catalog vocabularies	Publishing listings outward to enterprise catalogs and, where relevant, public registers	Export validates; a round-trip into a connected catalog preserves identifiers
Open table and sharing formats	Cross-platform product distribution without copying data	A product is consumed from a second engine through a share with no schema change
WCAG 2.2 AA	Every user-facing surface including both meshes and all motion	Automated axe suite plus a manual keyboard and screen-reader pass per release

Where a standard is immature — agent interoperability descriptors are the clear example today — the platform generates the artifact from its own manifest and treats the standard as an output format rather than an internal model. That keeps the canonical model stable while the ecosystem settles.

32. Design System & Front-End Engineering

32.1 Token Architecture

Three token tiers. Primitives are raw values; semantic tokens name intent; component tokens bind intent to a component. Application code references only semantic and component tokens, which is what makes white-labelling a configuration change rather than a refactor.

```
primitive   navy.900 #0B2C4D   teal.600 #117A8B   gold.600 #B98A1B   ink.900 #1F2A33
semantic    surface.canvas · surface.raised · text.primary · text.muted · accent.primary
            action.primary · status.healthy · status.watch · status.risk · focus.ring
component   card.radius · card.shadow.rest · card.shadow.hover · ring.quality.track
            ribbon.speed.row1 · constellation.node.max · theatre.type.cps

PRODUCT_NAME, wordmark asset and the primitive palette are tenant theme configuration.
CI asserts that no hex value and no brand string appears outside the token definition files.
```

32.2 Component Inventory

Group	Components
Identity and status	Quality ring · freshness pill · certification mark · sensitivity chip · autonomy badge · trust banner · confidence chip
Discovery	Universal search · facet rail · product card · agent card · compare matrix · empty state with demand CTA
Detail	Persistent asset header · tab shell · schema table · contract viewer with diff · endpoint block with copy and test-access · adoption charts · value case panel
Conversation	Demo console shell · streaming answer block · citation chip · tool-trace rail · feedback control · out-of-scope handoff card
Graph	Mesh canvas · node peek card · edge rationale popover · blast-radius control · accessible edge table
Workflow	Request form · policy evaluation preview · approval card · SLA clock · entitlement row · backlog item with votes
Motion	Motion controller · ribbon track · constellation renderer · answer theatre · counter · activity ticker

32.3 Engineering Rules

- **Server components by default.** The landing page and every catalog page render on the server with no client-side data dependency for first paint. Client components exist only where interaction requires them.
- **Every component ships five states**: loading, empty, error, partial-permission and populated. Partial-permission is the one teams forget and the one that matters most here, because a consumer routinely sees an asset they cannot yet query.
- **Skeletons are for in-app surfaces only.** The marketing landing page never renders skeletons; it renders cached content with an as-of stamp or it omits the band.
- **Responsive contract**: three breakpoints (≥1440, ≥1024, <768) with defined degradations for the ribbon, constellation and mesh explorer at each. Mobile is a first-class read surface for approvals and listings, not an afterthought.
- **Internationalisation from day one**: externalised copy, locale-aware number, date and currency formatting, layout verified at 35% string expansion, and RTL mirroring for layout and mesh direction.

- **Storybook plus visual regression** on every component in every state and both motion modes. A visual diff is a review conversation, not an automatic failure.

33. Testing, Environments & Release Engineering

33.1 Test Strategy

Layer	Scope	Bar
Unit	Scoring arithmetic, rubric resolution, mesh edge maths, deflection model, policy evaluation	Every rubric-driven calculation has a table-driven test asserting the value changes when the rubric changes
Contract	Generated OpenAPI, MCP servers and SDKs against the manifests that produced them	Regeneration produces no diff; a hand edit to a generated file fails the build
Golden fixture	Score reproducibility: a pinned evidence set plus a pinned rubric version reproduces a prior composite exactly	Byte-identical re-score, asserted for at least three historical snapshots
Publish gate	The full product and agent gates implemented as executable tests	An agent with four demo exchanges, a missing limitation statement or an uncited KPI is rejected with a precise message
Kill test	Connector service principals attempting writes to every supported platform	Every attempt fails; the test runs against a real sandbox account, not a mock
Security	Entitlement escalation, purpose bypass, injection through retrieved content, cross-tenant access	Fails closed in every case; suite is versioned centrally and run on every release
End-to-end	The signature journey: search, demo, request, approve, provision, first query	Runs on every merge against seeded data, under 8 minutes
Performance	Landing page budgets, catalog search p95, mesh render, motion frame rate under CPU throttling	Budgets in Section 24 and 6.10 asserted; regression fails the build
Accessibility	Automated axe pass plus scripted keyboard and screen-reader journeys on the ten highest-traffic surfaces	Zero critical violations; motion suite runs in both motion modes
Load and soak	Catalog at 50k assets, 500 concurrent sessions, 50M daily telemetry events	Quarterly, and before any tenant onboarding above 10k assets
Chaos	Data plane unreachable, connector auth expiry, model provider outage, workflow engine restart mid-approval	Documented degradation holds; no approval is lost or double-applied

33.2 Environments and Release

- **Four environments**: local with seeded synthetic data, ephemeral preview per pull request, staging with a full seed catalog and a sandbox platform account, production per tenant.
- **Feature flags** for every user-visible change, with flags typed as release, experiment or operational and a maximum age before removal is enforced by a lint rule.
- **Migrations are forward-only and reversible-by-compensation.** Publication snapshots and audit tables are append-only and never migrated destructively.
- **Progressive delivery**: canary by tenant cohort for the platform, canary by traffic share for agents. Both are the same mechanism with different keys.
- **SLOs with error budgets**: catalog availability, request-to-provision latency, agent answer success rate and demo console availability. Budget exhaustion freezes feature work until reliability work restores it — the freeze is the point.

- **Runbooks are shipped artifacts**, linked from every alert, and exercised in a quarterly game day covering an upstream contract breach, an agent regression and an entitlement-provisioning failure.

34. Estate Bootstrap & Onboarding

The hardest week of a marketplace is the first one, when the shelf is empty. An empty marketplace teaches consumers not to return, and consumer attention does not come back cheaply. Bootstrap is therefore a designed capability, not an implementation detail.

34.1 Ninety-Day Path

Window	Activity	Outcome
Days 1–15	Connect platform and catalog connectors read-only. Harvest metadata, lineage, usage, cost and existing quality results. Where an AI-readiness assessment exists, import its findings as the starting evidence set rather than re-harvesting.	Estate inventory with usage ranking; the top 200 assets by query volume identified
Days 16–30	Candidate product detection: cluster heavily-queried tables and views by shared consumers, shared lineage and query co-occurrence into candidate products. Reconcile BI measures into a draft KPI register with conflicts surfaced.	A ranked candidate list with duplicate KPI definitions named and counted — usually the first genuinely uncomfortable artifact the programme produces
Days 31–60	Auto-draft manifests for the top candidates: proposed name, purpose, grain, columns with inherited classification, draft contract from observed behaviour, and observed adoption. Every inferred field carries confidence and rationale; stewards complete and approve.	Ten to fifteen products published with real contracts, real owners and real quality scores
Days 61–90	Publish the first three agents over the strongest products, each with its coverage map, scope and five curated exchanges. Open access request rails. Open the demand board.	A live marketplace with supply, demonstrable agents, governed access and a demand signal

34.2 Bootstrap Rules

- **Drafts are never auto-published.** Inference accelerates authoring; a human owner approves before anything reaches the shelf. A catalog full of machine-generated descriptions nobody has read is the failure mode this rule exists to prevent.
- **Observed behaviour seeds the contract.** Freshness targets and volume bands are proposed from measured history, so the first contract describes reality and the owner argues upward from evidence rather than inventing a number.
- **Adoption history is imported, not reset.** A product that has had 300 consumers for two years launches showing that, because a launch that displays zero adoption for an established asset destroys credibility on day one.
- **Sequence supply before demand.** The demand board opens only once there is enough supply for duplicate detection to be meaningful; otherwise the first hundred requests all read as novel and the highest-value control never fires.
- **Define the first agent by an existing painful question**, identified from the query log and from what analysts are asked most often — not by what is technically easiest to build.

35. Product Analytics & Success Metrics

35.1 Metric Tree

One north star, four input metrics, four guardrails. A metric that is neither an input nor a guardrail is a dashboard decoration and is not instrumented.

Level	Metric	Why this one
North star	Governed answers per week — questions answered from a certified data product, by an agent or by a consumer, under an approved entitlement	It moves only when discovery, trust, access and capability all work; no single team can fake it
Input 1	Time from discovery to first successful query	The friction metric. Directly attacked by policy evaluation and mechanical provisioning
Input 2	Share of questions answered by an agent rather than escalated to an analyst	The deflection metric and the basis of the value case
Input 3	Certified supply coverage — share of the top 200 most-queried assets that are published products with contracts	The supply metric; without it the other three plateau
Input 4	Duplicate-supply redirect rate at intake	The cost-avoidance metric and the clearest evidence of marketplace value to a platform leader
Guardrail 1	Answer acceptance rate and groundedness	Deflection achieved by confident wrong answers is negative value
Guardrail 2	Permission-denied rate and dormant-grant share	Detects both over-restriction and over-provisioning
Guardrail 3	Cost per accepted answer	Prevents adoption growth from outrunning unit economics
Guardrail 4	Contract breach rate and quality composite trend	Prevents growth on a decaying foundation

35.2 Activation Funnel

```
visit → search → view listing → run a demo exchange → request access → approved
      → first query → second query within 14 days → habitual (4+ weeks in 8)

Instrumented at every step with drop-off attribution. Two conversions matter most:
  demo-exchange run   → access request      (does trying convert to committing?)
  first query         → second query        (did the asset actually serve the need?)

A listing whose demo-to-request conversion sits far below peer listings is a content
defect — usually a weak capability statement or an unconvincing curated question set —
and is routed to the owner as a specific, actionable finding rather than a generic report.
```

35.3 Instrumentation Rules

- A single typed event taxonomy shared by portal, API and agent runtime, versioned like any other schema. Ad-hoc event names are rejected in code review.
- Every event carries actor, actor type (human or agent), asset, surface, purpose where applicable, and tenant. Purpose on the event is what makes access audit and adoption analysis the same dataset.

- Analytics events are never the source of truth for entitlement or audit; the audit log is. Analytics may be sampled, audit may not.
- No third-party analytics beacon receives asset names, KPI names, query text or question text. Product analytics runs first-party against the tenant’s own store.

36. Commercial Model & Packaging

Included because packaging decisions constrain architecture. Multi-tenancy, metering and the deployment topology all have to be right before the first customer, and they are expensive to retrofit.

Edition	Contents	Deployment	Buyer	Pricing shape
Platform	Full catalog, meshes, workflows, observability, value module, academy, connectors	Customer cloud or platform-native (Snowpark Container Services and equivalents)	Data platform leadership, enterprise architecture	Platform fee plus tiering on published assets and connected platforms
Agent tier	Agent registry, demo console, AgentOps evaluation and release governance, agent mesh	Add-on to Platform	CDAO and AI programme leadership	Per live agent plus metered answers, with demo tier excluded from metering
Assessment and onboarding	Estate bootstrap, candidate product detection, KPI reconciliation, first-wave authoring with the customer	Services engagement using the platform	Transformation lead, CDO	Fixed fee by estate size; credited against the first Platform year
Partner edition	White-labelled deployment with tenant-configured branding and weighting profiles	Partner-hosted or customer-hosted	Systems integrators and advisory firms	Partner licence plus per-end-customer tier

- **Metering must be defensible.** Answers, published assets and connected platforms are countable, auditable and visible to the customer in the same place the platform counts them. A meter the customer cannot verify becomes a renewal argument.
- **Do not meter what you want to encourage.** Demo-tier answers, academy sandbox usage and demand-board submissions are never metered, because each is a growth loop.
- **Land on supply, expand on agents.** The catalog is the entry point because it is the safer purchase; agents are the expansion because they carry the visible deflection value. The architecture supports either order, but the go-to-market assumes this one.
- **Branding is configuration.** The partner edition requires no fork: product name, wordmark, palette and rubric weighting profiles are tenant configuration, enforced by the token architecture in Section 32.

37. Build Plan for Coding Agents

The implementation is specification-first. The coding agent (Claude Code, Codex or equivalent) is driven by the build markdown and the manifests and the manifests in this document; generated platform artifacts are renderings of a tool-neutral source of truth and are header-stamped as auto-generated. Nobody edits a generated file.

37.1 Repository Layout

```
marketplace/
  manifests/
    products/      DP-*.yaml         # tool-neutral product + contract source of truth
    agents/        AG-*.yaml         # capability, coverage map, scope, demo exchanges
    kpis/          KPI-*.yaml        # one authoritative definition per KPI
    rubrics/       quality.yaml, ranking.yaml, mesh.yaml, demand.yaml, value.yaml
    taxonomies/    industry.yaml, domain.yaml, archetype.yaml, purpose.yaml
  generated/       (DO NOT EDIT — header-stamped)
    sql/           semantic views, governed views, DMF attachments
    mcp/           one server definition per product
    openapi/       api spec + generated SDKs
    ddl/           canonical model migrations
  services/        catalog, search, mesh, workflow, entitlement, quality,
                   observability, value, academy, agent-registry, demo-runner, intake
  portal/          Next.js app
  connectors/      one package per platform and per governance/DQ tool
  seed/            demo-tier synthetic data generators + golden answers
  tests/           golden fixtures, publish-gate tests, kill tests, a11y suite
  skills/          SKILL.md instruction files driving agent-based build steps
```

37.2 Build Sequence

Step	Deliverable	Done when
1	Canonical model, migrations and rubric tables; manifest schema and validator	A malformed manifest fails validation with a precise error; no threshold exists outside a rubric table
2	Manifest-to-artifact generator: SQL views, MCP servers, OpenAPI, DDL	Regenerating from manifests produces byte-identical output; every generated file carries the auto-generated header
3	Snowflake connector end to end (metadata, lineage, DMF quality, usage, cost)	One real product harvested, scored and rendered; kill test proves the connector cannot write
4	Catalog service, hybrid search and ranking; data product catalog and detail pages	All eight product tabs render from real metadata; exact-name search never loses to a semantic neighbour
5	Quality engine and scoring snapshots	Re-running a score against a pinned rubric version reproduces it exactly; snapshots are append-only
6	Agent registry, coverage map, MCP tool binding, publish gate	An agent with four demo exchanges is rejected by the gate with a clear message
7	Demo runner and demo-tier synthetic data	All seed agents answer their five questions against synthetic data inside the latency budget
8	Workflow engine: access, enhancement, demand; policy	An approved request provisions a scoped platform role and writes an

Step	Deliverable	Done when
	evaluation and provisioning	immutable audit record
9	Mesh engine for both graphs, with confidence and rationale on every edge	Every rendered edge has a rationale; edges below 0.80 confidence are held for review
10	Observability plane and incident lifecycle	A simulated freshness breach raises an incident, notifies consumers and banners the listing
11	Value module and portfolio views; board-pack export	Exported deck and dashboard read from the same snapshot and cannot disagree
12	Landing experience, academy, admin console; accessibility and performance passes	LCP and WCAG 2.2 AA budgets met with the full seed catalog loaded

37.3 Instruction Files to Author First

- **product-manifest-authoring** — how to write a DP manifest: contract guarantees, column classification, quality rules as data, endpoint declarations, value case.
- **agent-manifest-authoring** — coverage map construction, scope minimization, demo exchange curation, evaluation set design, out-of-scope statement.
- **generator-conventions** — header stamping, determinism, no hand edits, regeneration test.
- **publish-gate** — the full gate checklist for both asset classes, implemented as executable tests rather than documentation.
- **mesh-computation** — edge derivation, weighting, confidence and rationale requirements, review routing below threshold.
- **demo-data-generation** — synthetic dataset generation preserving distribution, seasonality, referential integrity and cardinality, with no real subject data.

37.4 Non-Negotiable Engineering Constraints

- CI asserts that no numeric threshold, weight or grade band appears in application source; all live in rubric tables.
- CI asserts that connector service principals cannot write to any customer platform (kill test).
- CI asserts that regenerating all artifacts from manifests produces no diff.
- No row-data access path exists in the default build; the sampling module is a separate, permission-gated package with per-dataset authorization logging.
- Every LLM-generated description, mesh rationale, duplicate match and value estimate stores its prompt, inputs, confidence and reviewer decision.
- Publication snapshots are append-only; no code path updates a published snapshot in place.

38. Delivery Roadmap & Exit Criteria

Phase	Scope	Exit criteria
P1 — Demo-ready core (weeks 1–10)	Canonical model, manifests, generators, Snowflake connector, product catalog and detail, agent registry, demo console, seed catalog of 15 products and 14 agents, landing experience	A full client demonstration runs end to end on the seed catalog; all 70 curated exchanges pass nightly validation; landing page meets its LCP and accessibility budgets
P2 — Governed request rails (weeks 11–18)	Access, enhancement and demand workflows; policy evaluation; mechanical provisioning; entitlement register; duplicate-supply detection	Median time from request to provisioned access under one business day for auto-approve and owner-approve paths across three design partners; every grant carries a bound purpose and expiry
P3 — Mesh & observability (weeks 19–26)	Both meshes with confidence and rationale; observability plane for products and agents; incident lifecycle; upstream trust banners	Both meshes render the full seed estate with a rationale on every edge; a simulated upstream breach propagates to the agent listing within five minutes
P4 — Value & scale (weeks 27–36)	Value module, portfolio views, board-pack export, academy, Databricks and Fabric connectors, multi-tenancy hardening	Every published asset carries a reviewed value case with stated assumptions; a second platform is harvested through the same canonical model with no schema change
P5 — Marketplace network (beyond)	Cross-domain federation, external and partner data products, agent-to-agent composition with reconciliation checks, anonymized adoption benchmarking across tenants	Compositional exposure check blocks a genuinely over-scoped agent chain in a live estate; benchmark cohort reaches sufficient participants to publish comparisons

Two decisions to settle before P1 starts

Demo-tier data strategy: fully synthetic generation for all 15 seed products (cleanest for client demos and regulated prospects, higher build cost) versus obfuscated extracts from a design partner (faster, but constrains who the demo can be shown to). The specification assumes fully synthetic.

Agent runtime posture for the reference build: Snowflake-native only for P1 (fastest path, strongest security-review story, narrower prospect fit) versus a runtime-abstracted registry from the start (broader fit, materially more P1 work). The specification assumes Snowflake-native with adapters deferred to P4.

39. Client Demonstration Narrative

A 20-minute demonstration structured so that every requirement in this specification is shown rather than described. Industry is selected on the front page in step one, so the same script runs for a bank, a hospital system or a retailer without modification.

Min	Beat	What is shown	Point being made
0–2	The shelf	Landing page; select the prospect's industry; live counters and featured products re-render for that vertical	This is a product, not a catalog — and it already speaks your industry
2–4	Find	Type a business question into universal search; results return a product and an agent ranked by KPI match, with quality and adoption badges visible before any click	Discovery is by question, not by table name
4–8	Watch it answer	Open the agent, run scripted questions 2 and 3; answer streams with a chart, the KPI definition, an as-of stamp and citations; open the right rail to show tool calls, rows scanned, latency and cost	Try before you buy — and every number is traceable
8–10	Trust	Jump to the source data product: quality score with its six dimensions, the contract with conformance history, and the incident record	Trust is rendered from telemetry, not asserted in a wiki
10–12	Consume	Endpoints tab: copy the MCP server definition and the SQL view; show the 403 response carrying the exact missing scope and a deep link to request it	Any compliant agent can consume this; dead ends do not exist
12–14	Request	File an access request; policy evaluation shows the approval path and SLA before submission; approve it live; show the provisioned role, bound purpose and expiry in the entitlement register	Governance is a rail, not a queue
14–16	The mesh	Open the data mesh; hover a node to show shared sources; switch to blast-radius mode on a source system; switch to the agent mesh and show a KPI-overlap consolidation candidate	This is the estate's shape — and where duplication is forming
16–18	Unmet demand	File a new-supply request; duplicate detection surfaces an existing product at 0.81 similarity and blocks progression pending review; then show the demand board with clustered themes	The highest-value control in the product turns build requests into reuse decisions
18–20	The case	Portfolio value view: adoption curves, deflected hours with the derivation shown, value-versus-cost quadrant, retirement candidates, and the generated board pack	The investment defends itself with arithmetic the audience can check

39.1 Demonstration Rules

- Never demonstrate on mocked answers. The demo tier runs the real agent against synthetic data and says so on screen — a prospect who discovers a mock loses trust in everything else shown.
- Show one failure deliberately: the out-of-scope refusal with a handoff to another agent. Boundaries are a feature, and demonstrating one is more persuasive than five successes.
- Keep the right rail open during at least one answer. Transparency of tool calls and cost is the strongest differentiator against a generic chat interface.
- End on the value view, not on a feature. The buying decision is a budget decision.

40. Risk Register & Anti-Requirements

The risks below are the ones that have killed comparable programmes. Each is paired with the specific mechanism in this specification that addresses it, so the mitigation is a design commitment rather than a project-management intention.

40.1 Risk Register

Risk	How it materialises	Mitigation in this specification
Empty-shelf launch	Marketplace opens with a dozen thin listings, consumers visit once and do not return	Section 34 bootstrap path; seed catalog of 15 products and 14 agents; publish gates that make a thin listing impossible rather than merely discouraged
Catalog decay	Listings drift out of date, quality scores go stale, owners disengage	Scores computed from telemetry rather than authored; value-case staleness rendered on the listing; dormant-asset and unreviewed-owner reporting in the portfolio view
Governance theatre	Approval workflows exist but everything is approved without reading	Approval cards carry the requester's existing access and prior denials; partial approval requires a reason; dormant grants are surfaced back to the approver who granted them
Agent trust collapse	One confidently wrong answer to an executive ends the programme	Groundedness gate at 100%; blocking evaluation suites; upstream trust banners; the console refuses rather than guesses; recorded traces validated nightly
KPI divergence	Two agents give different answers for the same measure	Section 28 registry with one definition per name, cross-agent reconciliation, and divergence treated as an incident against both agents
Security review stall	The AI programme halts because agent entitlement scope cannot be described	Distinct agent identities, intersection semantics, per-tool scopes, compositional exposure checks, and a complete access reconstruction for any answer
Cost surprise	Adoption succeeds and inference spend outruns the value case	Section 30 unit economics, cost classes on every tool, budgets with graduated throttling, cost per accepted answer as a standing guardrail metric
Shadow authorization	The marketplace slowly becomes the real access control plane and diverges from platform RBAC	All grants provisioned into and revoked from native RBAC; the marketplace stores no effective permissions of its own; reconciliation job detects drift
Rubric drift between domains	Domains quietly tune weights until scores are incomparable	Rubrics are central, versioned and forum-approved; tenant weighting profiles are explicit and displayed alongside every benchmark
Motion as decoration	The animated front page is beautiful, slow, and inaccessible	Section 6.10 budgets enforced in CI, transform-and-opacity-only rule, static equivalents for every ambient animation, global pause control
Demo-to-production gap	The demo works and the real deployment does not	Demo tier runs the same code path against synthetic data; no mocked answers; the same publish gates apply to seed assets as to customer assets
Standards lock-in	A proprietary interface becomes the reason a customer cannot leave, and then the reason they never buy	Section 31 conformance tests; manifests and lineage exportable; products consumable from a second engine without schema change

40.2 Anti-Requirements

Things this product will deliberately not do, recorded so that they are re-litigated with intent rather than added quietly under delivery pressure.

- **No auto-publishing of inferred content.** Inference drafts; humans approve. This will be challenged every time the backlog is long, and the answer does not change.
- **No effective-permission store.** The marketplace will not cache or compute what a user can see. It requests, records and reconciles; the platform decides.
- **No row-level data in the default build.** Preview and sampling remain a separate, permission-gated package with per-dataset authorization logging.
- **No mocked demo answers, ever**, including in sales-led demonstrations. A recorded real trace is acceptable and labelled; a fabricated one is not.
- **No score without a rubric version.** A number on a screen that cannot be reproduced against a pinned rubric is not shipped.
- **No agent without an out-of-scope statement.** A boundary the owner cannot articulate is a boundary the platform cannot enforce.
- **No metric in the product analytics layer that is neither an input nor a guardrail** in the Section 35 tree.
- **No brand string or hex value in application source.** Theming is configuration; a fork for a partner deployment is a defect.

Appendix A — Requirement Traceability

Every requirement in the original brief, and every addition from the architecture review, mapped to the section that specifies it.

Requirement	Specified in
Browse marketplace for domain-specific data products	§5 IA · §7 Catalog · §25 Seed catalog
Browse AI agents and see which KPIs each can answer	§10 Agent catalog · §11.2 KPI coverage map · §26 Agent register · §28 KPI registry
Request access for data products and agents	§13 Access request, policy evaluation, mechanical provisioning
Request enhancements on a data product	§14 Enhancement workflow and public backlog
Request new data products and agents	§15 Demand intake, duplicate detection, demand board
Consumption patterns, DQ score, owner, contracts, MCP/AI endpoints per product	§8.4 Quality · §8.5 Contract · §8.8 Consumption · §9 Endpoints including MCP
Clear agent description and how it helps business analysis	§11.1 Overview with required business-value and out-of-scope blocks
At least five demo questions and answers per agent	§12 Demo console · §26.3 Publish gate · §27 Seventy scripted exchanges
Usage, adoption and value case for every asset	§22 Value realization · §8.9 · §11.7 · §25.3 · §26.2 · §30 Unit economics
Mesh for data products linked by shared sources	§16 Data product mesh
Mesh for AI agents linking similar agents	§17 Agent mesh
Data model	§21 Canonical model, relationships, physical schema, invariants
Academy	§20 Learning paths, sandboxes, certification
Observability for products and agents	§19 Signals and incident lifecycle · §29 AgentOps
Coverage of all major industries, 10+ products and 10+ agents	§25 Fifteen products · §26 Fourteen agents across nine industries; taxonomy supports twelve
Enterprise grade and client-demo ready	§24 NFR, security, compliance · §33 Testing and release · §38 Roadmap · §39 Demo narrative
Best-in-class front page	§6.1–6.4 Landing experience, visual system, ten bands, budgets
Moving data product and agent visuals on the front page	§6.5 Motion system · §6.6 Living data product ribbon · §6.7 Agent constellation · §6.8 Live answer theatre · §6.9 Activity ticker · §6.10 Motion budget
Unbranded, white-labelable product identity	Title block · §32.1 Token architecture · §36 Partner edition
Single authoritative KPI definitions	§28 Certified KPI registry and divergence detection
Agent evaluation, safety and release governance	§29 AgentOps
Cost control and unit economics	§30 Cost governance and FinOps
Open standards and interoperability	§31 Standards conformance tests

Requirement	Specified in
Design system and front-end engineering discipline	§32
Test strategy, environments, SLOs and release process	§33
Getting from empty catalog to live marketplace	§34 Estate bootstrap, ninety-day path
Product success measurement	§35 Metric tree, activation funnel, instrumentation
Packaging and commercial constraints on architecture	§36
Known risks and explicit non-goals	§40 Risk register and anti-requirements

End of specification. The companion build markdown renders this specification as an executable task plan for a coding agent; manifests, rubric YAML and golden-answer fixtures are generated from both.